

A LONG-TERM INTERNSHIP REPORT

On

Machine Learning Using Python

Submitted To Department Of Computer Science

By

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III BCA

Under The Esteemed Guidance Of

Ms.K.BHAVANA SRAVANTHI

LECTURE

Department Of Computer Science



**ADITYA DEGREE COLLEGE
FOR WOMENS GOPALAPATNAM**



ADHOC NETWORK

Tech Company

CERTIFICATE Of Internship

Proudly Presented To:

DEVAGUPTAPU SRAVANI

a student of Aditya Degree College, has successfully
completed Machine Learning Using Python
Internship at Adhoc Network Tech Company from
February 08, 2024 to April 20, 2024.



Project Coordinator

ADITYA DEGREE COLLEGE

Department of Computer Science



CERTIFICATE

This is to certify that The Long Term Internship entitled,

“Machine Learning Using Python” is a bonified work of **Sravani Devaguptapu**, bearing **121127306037**, III **BCA**, submitted to the Department of Computer Science, Aditya Degree College for women, Gopalapatnam for the academic year 2021-2024.

Internship Guide

Miss.N. SANDHYA , B.Tech

Project Coordinator

Adhoc Network Tech Company

Head of the Department

Ms.K Bhavana Sravanthi

External Examiner Principal

ADITYA DEGREE COLLEGE

Department of Computer Science



DECLARATION BY THE STUDENT

I hereby declare that the work described in this Long Term Internship, entitled “**Machine Learning Using Python**” which is being submitted by me in partial fulfillment of the requirements for the award of degree of **Bachelor of Computer Applications (BCA)** from Aditya Degree College for women, Gopalapatnam under the guidance of **N.SANDHYA , Project Coordinator of Adhoc Network Tech Company, India and Canada.**

Place: Gopalapatnam

Date: 20-04-2024

ADITYA DEGREE COLLEGE

Department of Computer Science



CERTIFICATE FROM THE SUPERVISOR

—This is to certify that the Long Term Internship entitled, "**Machine Learning Using Python**", that is being submitted by **Sravani Devaguptapu** bearing **121127306037, III BCA** , which is being submitted by me in partial fulfillment of the requirements for the award of degree of Bachelor of Computer Applications from **Aditya Degree College for women**, bonafied work carried out by him under my guidance and Supervision.

MS.M.RADHA

ACKNOWLEDGEMENT

No endeavour is completed without the valuable support of others. I would like to take this opportunity to extend my sincere gratitude to all those who have contributed to the successful completion of this Short-Term Internship Project Report.

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It is privilege to thank **Dr. N. SUGUNA REDDY**, Secretary Madam, Aditya group of institutions for providing Short-Term Internship Project Report from Adhoc.

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Finally I thank all the faculty members of our Department who contributed their valuable suggestions in completion of Long-Term Internship report and I also put my sincere thanks to My Parents who stood with me during the whole Long-Term Internship.

- *Sravani Devaguptapu*

ABOUT ADITYA DEGREE COLLEGES



Dr.N.SESHA REDDY
CHAIRMAN



Dr.SUGUNA REDDY
SECRETARY



Dr.B.E.V.L. NAIDU
ACADEMIC DIRECTOR

Aditya Degree colleges are the precious gifts presented to the twin Godavari Districts by ADITYA. Educational group ADITYA Degree College which was established in 1998 in Kakinada fulfilled the hopes and aspirations of many graduates and had been acclaimed as the best degree college under Andhra University. Encouraged by the 100% result in 2003, ADITYA added several feathers to its cap by launching Degree Colleges in Rajahmundry in 2003, in Vizag and Palakol in 2005 and in Tatipaka in 2006.

Needless to say, in the present scenario girls excel more than boys in education and they give tough competition to boys. Owing to their diffidence and inhibition, girls find difficulty in expressing their doubts in a classroom of Co-ed College. Moreover, parents have many objections in sending their daughters to a co-ed college. Realizing this, ADITYA successfully leads Degree colleges for girls to prove their talents in curricular, co-curricular and extra-curricular activities. ADITYA started P.G College also for Women to encourage them for higher education.

VISION

To provide inclusive education with innovative methods and strenuous efforts for inculcating human values, professionalism and scientific instillation in the realm of Degree Education to all sections of students irrespective of race, region and religion with special focus to stand independently and to emerge as centre for Research and Development.

MISSION

To provide ample scope for multifaceted development of local youth. To provide quality higher education to student community.

To Recruit Highly Qualified and Experienced Faculty to provide Quality Education.

ABOUT ADHOC NETWORK TECH COMPANY - INDIA,CANADA

Adhoc Network is started in the year 2020 at Visakhapatnam by an young Women entrepreneur Miss. Devika Pakruthi Founder & CEO with an intention to provide employment opportunities to the youth and also to impart the best quality training and practical exposure to the Students which enhances their employability Skills. Her journey started and collapsed with the wide spread of Covid-19 but her determination and aspirations made her journey more futuristic and she never gave up the thought to GIVE-UP. This is where Devika made her dreams come true and alive. She never expected with a sole objective of making Profit but her determination to impart the quality training made her to reach the peaks of success at the young age.

Adhoc Network-we are proud to have been awarded a Hatrick of Awards. This Achievement is a testament to our commitment to Excellence and Innovation in the Software development Best heading company in the market.

Devika Pakruthi, a name synonymous with innovation, empowerment, and success. As the proud recipient of the Young Entrepreneur Award, Best Women- Led Startup Award, Women Rising Star of the Year Award and Youngest CEO of the Year Award, Devika Pakruthi has etched her name into the annals of contemporary business history.

Vision, Mission and values of the

Organization:- VISION:-

Due vision is to be a leading global provider of interactive and reliable Software Solution empowering business to Thrive in the digital age.

MISSION:-

Our mission is to develop cutting-edge Software Solutions that Solve compare business challenges, enhance Operations efficiency and drive Sustainable growth for our client we strive to deliver exceptional value by leveraging emerging technologies, fostering Strategic partnerships and maintaining a Customer, Centric approach Values

- Innovation
- Excellence
- Collaboration
- Integrity
- Customer Centricity
- Continuous Learning

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PYTHON PROGRAMS

Experiment 1

Title: Write a program of hello world.

Source code

```
print("Hello World!")
```

Output



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\SRAVANI\Desktop\webtech> & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '59757' '--' 'c:\Users\SRAVANI\Desktop\Python\H.py'
Hello World!
PS C:\Users\SRAVANI\Desktop\webtech>
```

Experiment 2

Title: Write a program on arithmetic operations.

Source code

```
num1=float(input("Enter the first number"))

op=input("Enter operators(+,-,*,/):")

num2=float(input("Enter the second number"))

if op=="+":

    print(num1+num2)

elif op=="-":

    print(num1-num2)

elif op=="*":

    print(num1*num2)

elif op=="/":

    if num2!=0:

        print(num1/num2)

    else:

        print("error:Division by Zero")

else:

    print("Invalid Operator")
```

Output:

```
PS C:\Users\SRAVANI\Desktop\webtech> c::; cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundle\libs\debugpy\adapter\...\debugpy\launcher' '59780' '--' 'c:\Users\SRAVANI\Desktop\Python\Arithmetic.py'
Enter the first number2
Enter operators(+,-,*,/):+
Enter the second number3
5.0
```

Experiment 3

Title: Write a program to simple interest.

Source code

```
principal=float(input("Enter the principal amount:"))

rate=float(input("Enter the anual intreast rate(%):"))

time=float(input("Enter the period in year"))

Simple_intrest=(principal*rate*time)/100

print("Simple Intrest:",Simple_intrest)
```

Output:

```
PS C:\Users\SRAVANI\Desktop\webtech> ^C
PS C:\Users\SRAVANI\Desktop\webtech> c::; cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '59793' '--' 'c:\Users\SRAVANI\Desktop\Python\Simple.py'
Enter the principal amount:5000
Enter the anual intreast rate(%):1
Enter the period in year2
Simple Intrest: 100.0
```

Experiment- 4

Title: Write a program to Fibonacci series.

Source code

```
n=int(input("Enter the number of terms:"))

n1,n2=0,1

if n<=0:

    print("please enter a possitive number")

elif n==1:

    print("Fibonacci sequence upto the given num",n, "is:",n1)

else:

    print("Fibonacci sequence is:")

    print(n1,end=" ")

    print(n2,end=" ")

    for i in range (2,n):

        next_term=n1+n2

        print(next_term,end=" ")

        n1,n2=n2,next_term
```

Output:

```
PS C:\Users\SRAVANI\Desktop\webtech> ^C
PS C:\Users\SRAVANI\Desktop\webtech> c:: cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode
\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '59805' '--' 'c:\Users\SRAVANI\Desktop\Python\
Fibonacci.py'
Enter the number of terms:5
Fibonacci sequence is:
0 1 1 2 3
```

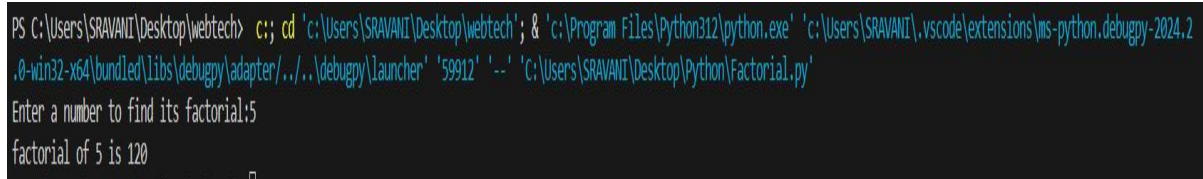

Experiment 5

Title: Write a program on factorial.

Source code

```
def factorial(n):  
  
    if n==0:  
  
        return 1  
  
    else:  
  
        return n*factorial(n-1)  
  
num=int(input("Enter a number to find its factorial:"))  
  
print("factorial of",num,"is",factorial(num))
```

Output:



```
PS C:\Users\SRAVANI\Desktop\webtech> c:: cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2  
.0-win32-x64\bundle\libs\debugpy\adapter\..\..\debugpy\launcher' '59912' '--' 'c:\Users\SRAVANI\Desktop\Python\Factorial.py'  
Enter a number to find its factorial:5  
factorial of 5 is 120
```

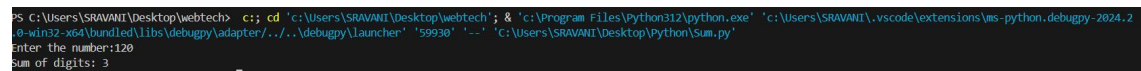
Experiment 6

Title: Write a program to calculate sum of digits of the given number

Source code

```
def sum_of_digits(n):  
  
    sum=0  
  
    while n>0:  
  
        rem=n%10  
  
        sum=sum+rem  
  
        n=n//10  
  
    return sum  
  
num=int(input("Enter the number:"))  
  
print("Sum of digits:",sum_of_digits(num))
```

Output:



```
PS C:\Users\SRAVANI\Desktop\webtech> cd "C:\Users\SRAVANI\Desktop\webtech"; & "C:\Program Files\Python312\python.exe" "C:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2  
0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher" "59930" "..." "C:\Users\SRAVANI\Desktop\Python\Sum.py"  
Enter the number:120  
Sum of digits: 3
```

Experiment 7

Title: Write a program to find the given number is palindrome or not

Source code

```
def is_palindrome(num):  
    return str(num)==str(num)[::-1]  
  
num=int(input("Enter a number:"))  
  
if is_palindrome(num):  
    print(num,"is a palindrome")  
  
else:  
    print(num,"is not a palindrome")
```

Output:

```
PS C:\Users\SRAVANI\Desktop\webtech> & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '49865' '--' 'C:\Users\SRAVANI\Desktop\Python\palindrome.py'  
Enter a number:121  
121 is a palindrome
```

Experiment 8

Title: Write a program to find the given year is leap year or not

Source code

```
OutPut:def is_leap_year(year):

    if(year%4==0 and year%100!=0)or(year%400==0):

        return True

    else:

        return False

year=int(input("enter a year:"))

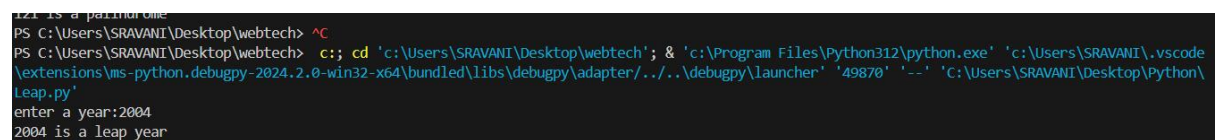
if is_leap_year(year):

    print(year,"is a leap year")

else:

    print(year,"is not a leap year")
```

Output



```
121 is a palindrome
PS C:\Users\SRAVANI\Desktop\webtech> ^C
PS C:\Users\SRAVANI\Desktop\webtech> c:; cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '49870' '--' 'c:\Users\SRAVANI\Desktop\Python\Leap.py'
enter a year:2004
2004 is a leap year
```

Experiment 9

Title: Write a program to find the given number is even or odd

Source code

```
num=int(input("enter a number:"))
```

```
if num%2==0:
```

```
    print(num,"is a even number")
```

```
else:
```

```
    print(num,"is a odd number")
```

OutPut:

```
PS C:\Users\SRAVANI\Desktop\webtech> c::; cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundled\libs\debugpy\adapter\..\debugpy\launcher' '49875' '--' 'C:\Users\SRAVANI\Desktop\Python\Odd.py'
enter a number:10
10 is a even number
```

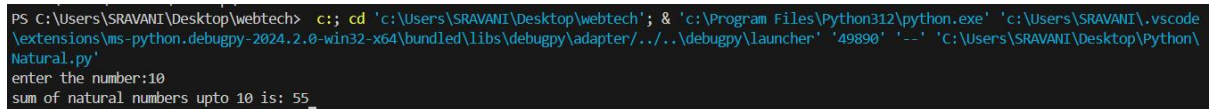
Experiment 10

Title: Write a program to find the sum of natural numbers upto a given number

Source code

```
def sum_of_natural_numbers(n):  
    return n*(n+1)//2  
  
num=int(input("enter the number:"))  
  
if num<0:  
    print("please enter a positive number")  
  
else:  
    print("sum of natural numbers upto",num,"is:",sum_of_natural_numbers(num))
```

OutPut:



```
PS C:\Users\SRAVANI\Desktop\webtech> cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '--' 'c:\Users\SRAVANI\Desktop\Python\Natural.py'  
enter the number:10  
sum of natural numbers upto 10 is: 55
```

Experiment 11

Title: Write a program to perform the Relational Operator

Source code

```
n1=int(input("enter a 1st number:"))

rop=input("enter a operator(<,<=,>,>=,==,!=):")

n2=int(input("enter a 2nd number:"))

if rop=="<":

    print(n1<n2)

elif rop=="<=":

    print(n1<=n2)

elif rop==">":

    print(n1>n2)

elif rop==">=":

    print(n1>=n2)

elif rop=="==":

    print(n1==n2)

elif rop=="!=":

    print(n1!=n2)

else:

    print("Invalid operator")
```

OutPut:

```
PS C:\Users\SRAVANI\Desktop\sravani\Human Stress Detection Based on Sleeping Habits> & 'c:\Users\SRAVANI\AppData\Local\Programs\Python\Python310\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.4.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '51148' '--' 'C:\Users\SRAVANI\Desktop\abc.py'
enter a 1st number:2
enter a operator(<,<=,>,>=,==,!=):!=
enter a 2nd number:67
Invalid operator
```


Experiment -12

Title: Write a program to implement a single python calculator using functions in while loop

Source code

```
def add(x,y):

    return x+y

def subtract(x,y):

    return x-y

def multiply(x,y):

    return x*y

def divide(x,y):

    if y==0:

        return "cannot divide by zero"

    else:

        return x/y

print("Select option :")

print("1.Add")

print("2.Subtract")

print("3.Multiply")

print("4.Divide")

while True:

    choice=input("Enter choice(1/2/3/4):")

    if choice in('1','2','3','4'):

        num1=int(input("Enter the 1st number :"))

        num2=int(input("Enter the 2nd number :"))
```

```

if choice=='1':

    print("Result:",add(num1,num2))

elif choice=='2':

    print("Result:",subtract(num1,num2))

elif choice=='3':

    print("Result:",multiply(num1,num2))

elif choice=='4':

    print("Result:",divide(num1,num2))

else :

    print("Invalid Input")

again=input("Do you want to perform another calculation?(Yes/No):")

if again.lower()!="yes":

    break

```

OutPut:

```

PS C:\Users\SRAVANI\Desktop\webtech> ^C
PS C:\Users\SRAVANI\Desktop\webtech> c:: cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundle\libs\debugpy\adapter\..\..\debugpy\launcher' '49900' '--' 'C:\Users\SRAVANI\Desktop\Python\Calculator.py'
Select option :
1.Add
2.Subtract
3.Multiply
4.Divide
Enter choice(1/2/3/4):4
Enter the 1st number :2
Enter the 2nd number :1
Result: 2.0
Do you want to perform another calculation?(Yes/No):Yes
Enter choice(1/2/3/4):2
Enter the 1st number :3
Enter the 2nd number :4
Result: -1
Do you want to perform another calculation?(Yes/No):N0

```

Experiment -13

Title: Write a program to perform addition of numbers using functions

Source code

```
num1=int(input("enter the first number:"))

num2=int(input("Enter the second number:"))

def add(n1,n2):

    return n1+n2

print("Addition of",num1,"and",num2,"is:",add(num1,num2))
```

OutPut:

```
PS C:\Users\SRAVANI\Desktop\webtech> ^C
PS C:\Users\SRAVANI\Desktop\webtech> c;; cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundle\libs\debugpy\adapter\..\..\debugpy\launcher' '49905' '--' 'c:\Users\SRAVANI\Desktop\Python\Addition.py'
enter the first number:10
Enter the second number:2
Addition of 10 and 2 is: 12
```

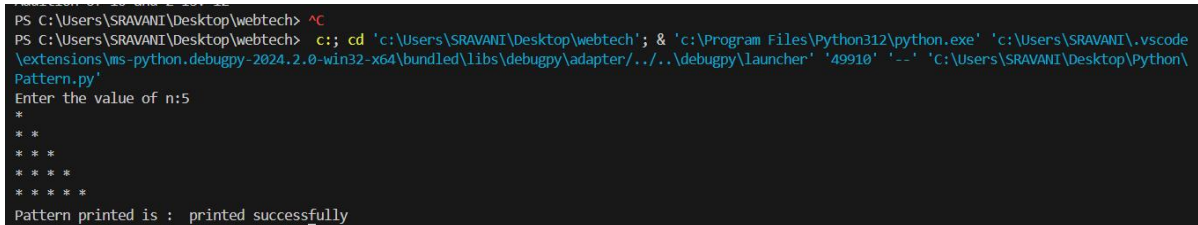
Experiment-14

Title: Write a program to print start in increasing triangular format

Source code

```
def pattern(n):  
  
    for i in range(1,n+1):  
  
        for j in range(0,i):  
  
            print("*",end=" ")  
  
        print()  
  
    return "printed successfully"  
  
n=int(input("Enter the value of n:"))  
  
print("Pattern printed is : ",pattern(n))
```

OutPut:



```
PS C:\Users\SRAVANI\Desktop\webtech> ^C  
PS C:\Users\SRAVANI\Desktop\webtech> c:: cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundle\libs\debugpy\adapter\...\debugpy\launcher' '49910' '--' 'c:\Users\SRAVANI\Desktop\Python\Pattern.py'  
Enter the value of n:5  
*  
* *  
* * *  
* * * *  
* * * * *  
Pattern printed is : printed successfully
```

Experiment -15

Title: Write a program to find the biggest of three numbers.

Source code

```
num1=float(input("Enter the first number:"))

num2=float(input("Enter the second number:"))

num3=float(input("Enter the third number:"))

if num1>=num2 and num1>=num3:

    greatest=num1

elif num2>=num1 and num2>=num3:

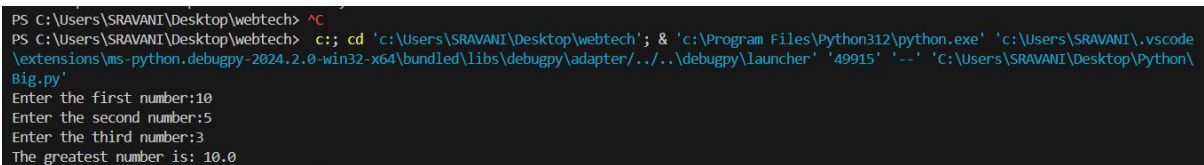
    greatest=num2

else:

    greatest=num3

print("The greatest number is:",greatest)
```

Output:



```
PS C:\Users\SRAVANI\Desktop\webtech> ^C
PS C:\Users\SRAVANI\Desktop\webtech> c:; cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundle\libs\debugpy\adapter\..\..\debugpy\launcher' '49915' '--' 'c:\Users\SRAVANI\Desktop\Python\Big.py'
Enter the first number:10
Enter the second number:5
Enter the third number:3
The greatest number is: 10.0
```

Experiment -16

Title: Write a program to implement logical operators.

```
x=int(input("Enter the value of x:"))

y=int(input("Enter the value of y:"))

print(x,">0 and",x,"<10 is:",x>0 and x<10)

print(x,">0 and",y,">10 is:",x>0 and y>10)

print(x,">10 or",y,">10 is:",x>10 or y>10)

print(x,"%2==0 and",y,"%2==0 is:",x%2==0 and y%2==0)

print("not(",x,"+",y,">15)is:",not(x+y)>15)
```

Output:

```
PS C:\Users\SRAVANI\Desktop\webtech> ^C
PS C:\Users\SRAVANI\Desktop\webtech> c:: cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '49920' '--' 'C:\Users\SRAVANI\Desktop\Python\logical.py'
Enter the value of x:10
Enter the value of y:5
10 >0 and 10 <10 is: False
10 >0 and 5 >10 is: False
10 >10 or 5 >10 is: False
10 %2==0 and 5 %2==0 is: False
not( 10 + 5 >15)is: True
```

Experiment -17

Title: Write a program to implement bitwise operators.

Source code

```
x=int(input("Enter the value of x:"))
y=int(input("Enter the value of y:"))

result_and=x&y

print("Bitwise AND of",x,"and",y,"is:",result_and)

result_or=x|y

print("Bitwise OR of",x,"and",y,"is:",result_or)

result_xor=x^y

print("Bitwise XOR of",x,"and",y,"is:",result_xor)

result_not_x=~x

print("Bitwise NOT of",x,"is:",result_not_x)

shift_bits=int(input("Enter the number of bits to left shift x by:"))

result_left_shift=x<<shift_bits

print("Bitwise left shift of",x,"by",shift_bits,"bits is:",result_left_shift)

shift_bits=int(input("Enter the number of bits to right shift y by:"))

result_right_shift=y>>shift_bits

print("Bitwise right shift of",y,"by",shift_bits,"bits is:",result_right_shift)
```

Output:

```
PS C:\Users\SRAVANI\Desktop\webtech> ^C
PS C:\Users\SRAVANI\Desktop\webtech> c;; cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '49925' '--' 'C:\Users\SRAVANI\Desktop\Python\Bitwise.py'
Enter the value of x:5
Enter the value of y:2
Bitwise AND of 5 and 2 is: 0
Bitwise OR of 5 and 2 is: 7
Bitwise XOR of 5 and 2 is: 7
Bitwise NOT of 5 is: -6
Enter the number of bits to left shift x by:10
Bitwise left shift of 5 by 10 bits is: 5120
Enter the number of bits to right shift y by:20
Bitwise right shift of 2 by 20 bits is: 0
```


Experiment-18

Title: Write a program to implement bitwise assignment operators.

Source code

```
X=int(input("enter the initial value of X:"))

print("initial value of X:",X)

X+=5

print("value of X after addition assignment is:",X)

X-=3

print("value of X after subtraction assignment is:",X)

X*=2

print("value of X after multiplication assignment is:",X)

X/=4

print("value of X after division assignment is:",X)

X%=3

print("value of X after modulus assignment is:",X)

X**=2

print("value of X after exponentation assignment is:",X)

X//=2

print("value of X after floor division assignment is:",X)
```

Output:

```
PS C:\Users\SRAVANI\Desktop\webtech> ^C
PS C:\Users\SRAVANI\Desktop\webtech> c;; cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '49930' '--' 'c:\Users\SRAVANI\Desktop\Python\Assignment.py'
enter the initial value of X:10
initial value of X: 10
value of X after addition assignment is: 15
value of X after subtraction assignment is: 12
value of X after multiplication assignment is: 24
value of X after division assignment is: 6.0
value of X after modulus assignment is: 0.0
value of X after exponentation assignment is: 0.0
value of X after floor division assignment is: 0.0
```

Experiment -19

Title: Write a program to implement perfect squares between 1 and 50.

Source code

```
def is_perfect_square(num):  
  
    return int(num**0.5)**2==num  
  
for i in range(1,51):  
  
    if is_perfect_square(i):  
  
        print(i,end=" ")
```

Output:

```
PS C:\Users\SRAVANI\Desktop\webtech> ^C  
PS C:\Users\SRAVANI\Desktop\webtech> c::; cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode  
\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundle\libs\debugpy\adapter\..\..\debugpy\launcher' '49935' '--' 'c:\Users\SRAVANI\Desktop\Python\  
Perfect.py'  
1 4 9 16 25 36 49
```

Experiment -20

Title: Write a program to implement type conversions in python

Source code

```
int_num=int(input("Enter an integer value:"))

float_num=float(int_num)

print("Inter to float conversion is:",float_num)

float_num=float(input("Enter a flaot value:"))

int_num=int(float_num)

print("float to integer coversion is:",int_num)

str_num=input("Enter a string representing a integer value:")

int_from_str=int(str_num)

print("string to integer conversion is:",int_from_str)

str_float=input("Enter a string representing a float number:")

float_from_str=float(str_float)

print("float string conversion is:",float_from_str)

float_str=float(input("Enter a float number:"))

str_from_float=str(float_str)

print("Float to string conversion is:",str_from_float)
```

output:

```
Enter a string representing a integer value:5
PS C:\Users\SRAVANI\Desktop\webtech> c:: cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundle\libs\debugpy\adapter\..\..\debugpy\launcher' '49945' '---' 'C:\Users\SRAVANI\Desktop\Python\Type conversion.py'
Enter an integer value:5
Inter to float conversion is: 5.0
Enter a flaot value:1.5
float to integer coversion is: 1
Enter a string representing a integer value:12
string to integer conversion is: 12
Enter a string representing a float number:11.2
float string conversion is: 11.2
Enter a float number:1.8
Float to string conversion is: 1.8
```

Experiment -21

Title: Write a program to implement break, continue and pass statements in python

Source code

```
my_list=[6,7,8,9,10]

print("Demonstrating 'break statement':")

for num in my_list:

    if num==9:

        break

    print(num)

print("Outside loop")

print("\nDemonstrating 'Continue statement':")

for num in my_list:

    if num%2==0:

        continue

    print(num)

print("Outside loop")

print("\nDemonstrating 'pass statement':")

for num in my_list:

    if num==7:

        pass

    else:

        print(num)

print("Outside loop")
```

OutPut:

```
PS C:\Users\SRAVANI\Desktop\webtech> ^C
PS C:\Users\SRAVANI\Desktop\webtech> c::; cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundle\libs\debugpy\adapter\..\..\debugpy\launcher' '49950' '---' 'C:\Users\SRAVANI\Desktop\Python\Break.py'
Demonstrating 'break statement':
6
7
8
Outside loop

Demonstrating 'Continue statement':
7
9
Outside loop

Demonstrating 'pass statement':
6
8
9
10
Outside loop
```

Experiment -22

Title: Write a program to implement mathematical functions in python

Source code

```
import math

num_sqrt=float(input("\nenter a number to find it's square root:"))

print("Square root of",num_sqrt,"is:",math.sqrt(num_sqrt))

num_factorial = int(input("\nEnter a number to find its factorial:"))

print("Factorial of",num_factorial,"is:",math.factorial(num_factorial))

num_abs=float(input("\n Enter a floating point number to find its absolute value:"))

print("Absolute value of",num_abs,"is:",math.fabs(num_abs) )

base= float(input("\nEnter the base to find exponential:"))

exponent=float(input("Enter the exponent of an exponential:"))

print(base,"raised to The power of",exponent,"is:",math.pow(base,exponent))

num_log=float(input("\nEnter a number to find its logarithm:"))

base_log=float(input("\nEnter the base of the logarithms:"))

print("Logarithms of",num_log,"to the base",base_log,"is:",math.log(num_log,base_log))

num1_gcd=int(input("\nEnter the first number:"))

num2_gcd=int(input("Enter the second number:"))

print("GCD of",num1_gcd,"and",num2_gcd,"is:",math.gcd(num1_gcd,num2_gcd))

print("\nValue of Euler's number(e):",math.e)
```

OutPut:

```
PS C:\Users\SRAVANI\Desktop\webtech> ^C
PS C:\Users\SRAVANI\Desktop\webtech> c:: cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundle\libs\debugpy\adapter\..\..\debugpy\launcher' '49955' '--' 'C:\Users\SRAVANI\Desktop\Python\Math.py'

enter a number to find it's square root:4
Square root of 4.0 is: 2.0

Enter a number to find its factorial:10
Factorial of 10 is: 3628800

Enter a floating point number to find its absolute value:1.5
Absolute value of 1.5 is: 1.5

Enter the base to find exponential:5
Enter the exponent of an exponential:2
5.0 raised to The power of 2.0 is: 25.0

Enter a number to find its logarithm:10

Enter the base of the logarithms:5
Logarithms of 10.0 to the base 5.0 is: 1.4306765580733933

Enter the first number:10
Enter the second number:5
GCD of 10 and 5 is: 5

Value of Euler's number(e): 2.718281828459045
```

Experiment -23

Title: Write a program to implement formatted string in python

Source code

```
name="Sravani"

age="20"

print(f"My name is {name} and I am {age}years old.")
```

Out Put:



```
PS C:\Users\SRAVANI\Desktop\webtech> ^C
PS C:\Users\SRAVANI\Desktop\webtech> c::; cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundle\libs\debugpy\adapter\..\..\debugpy\launcher' '49960' '--' 'C:\Users\SRAVANI\Desktop\Python\String.py'
My name is Sravani and I am 20years old.
```


Experiment -24

Title: Write a program to implement enumerate function in python

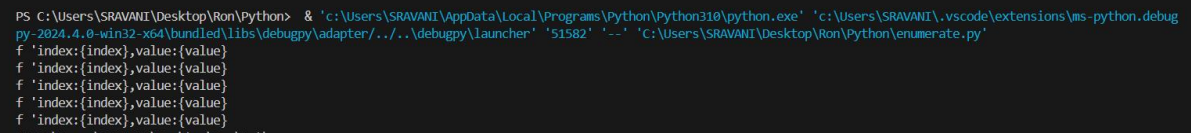
Source code

```
list=['Harry','Ron','23','Hagrid','08']

for index,value in enumerate(list):

    print(f'index:{index},value:{value}')
```

Output



```
PS C:\Users\SRAVANI\Desktop\Ron\Python> & 'c:\Users\SRAVANI\AppData\Local\Programs\Python\python310\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debug
py-2024.4.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '51582' '--' 'C:\Users\SRAVANI\Desktop\Ron\Python\enumerate.py'
f 'index:{index},value:{value}'
f 'index:{index},value:{value}'
f 'index:{index},value:{value}'
f 'index:{index},value:{value}'
f 'index:{index},value:{value}'
```

Experiment -25

Title: Write a program to achieve formatted string concept using string interpolation in python

Source code

```
todo_list = []

def add_task(task):

    todo_list.append(task)

    print (f'Task '{task}' added successfully to the to-do list.")

def view_tasks():

    if todo_list:

        print ("To-Do List:")

        for idx, task in enumerate (todo_list, start=1):

            print(f'{idx}. {task}')

    else:

        print ("To-Do List is empty.")

while True:

    print ("\n1. Add Task\n2. View Task\n3. Quit")

    choice = input ("Enter your choice: ")

    if choice == "1":

        task=input("enter the task:")

        add_task(task)

    elif choice == "2":

        view_tasks()

    elif choice == "3":
```

```
print ("Exiting...")
```

```
break
```

```
else:
```

```
print ("Invalid choice. Please try again.")
```

OutPut:

```
My name is Sravani and I am 20years old.  
PS C:\Users\SRAVANI\Desktop\webtech> c++; cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundle\libs\debugpy\adapter\..\..\debugpy\launcher' '49965' '--' 'c:\Users\SRAVANI\Desktop\Python\Interpolation.py'  
  
1. Add Task  
2. View Task  
3. Quit  
Enter your choice: 2  
To-Do List is empty.  
  
1. Add Task  
2. View Task  
3. Quit  
Enter your choice: 5  
Invalid choice. Please try again.  
  
1. Add Task  
2. View Task  
3. Quit  
Enter your choice: 12  
Invalid choice. Please try again.  
  
1. Add Task  
2. View Task  
3. Quit  
Enter your choice: 1  
enter the task:4  
Task '4' added successfully to the to-do list.
```

Experiment-26

Title: Write a program in python to search for an element in an array

Source code

```
from array import array

arr=array('i',[])

num_elements=int(input("Enter the length of an array"))

for i in range(num_elements):

    x=int(input("Enter the value:"))

    arr.append(x)

print("\n elements in an array:",arr)

ele_search=int(input("\n enter the search element:"))

index=0

found=False

for c in arr:

    if c==ele_search:

        print("Element found at index number:",index)

        found=True

        break

    index+=1

else:

    print("Element not found in the array using the traditional approach")

try:

    index=arr.index(ele_search)

    print("Using built in functions,element found at index:",index)
```

except ValueError:

print("element not found using built in function")

Output

```
PS C:\Users\SRAVANI\Desktop\webtech> c::; cd 'c:\Users\SRAVANI\Desktop\webtech'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.2.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '62694' '--' 'C:\Users\SRAVANI\Desktop\Python\from array import array.py'
Enter the length of an arrays
Enter the value:1

elements in an array: array('i', [1])

enter the search element:2
Enter the value:12

elements in an array: array('i', [1, 12])

enter the search element:3
Enter the value:1

elements in an array: array('i', [1, 12, 1])
```

HTML

1. Design a web page of Basic HTML Template

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>My Simple HTML Page</title>
```

```
</head>
```

```
<body>
```

```
  <header>
```

```
    <h1>Welcome to My Simple HTML Page</h1>
```

```
  </header>
```

```
  <div>
```

```
    <p>This is a simple HTML page created for demonstration purposes.</p>
```

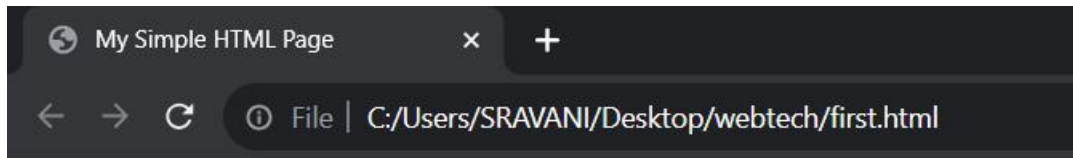
```
    <p>You can add more content here.</p>
```

```
  </div>
```

```
</body>
```

</html>

Output:



Welcome to My Simple HTML Page

This is a simple HTML page created for demonstration purposes.

You can add more content here.

2. Design a Simple Website Template with Navigation

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Example HTML Program</title>

</head>

<body>

<header>

    <h1>Welcome to My Website</h1>

    <nav>

        <ul>

            <li><a href="#home">Home</a></li>

            <li><a href="#about">About</a></li>

            <li><a href="#contact">Contact</a></li>

        </ul>

    </nav>

</header>

<div>
```


<h2>About Us</h2>

<p>HTML consists of a series of elements that enclose or wrap content to define its structure and meaning on a webpage. </p>

<p>Each element is represented by a tag, enclosed in angle brackets.. </p>

</div>

<footer>

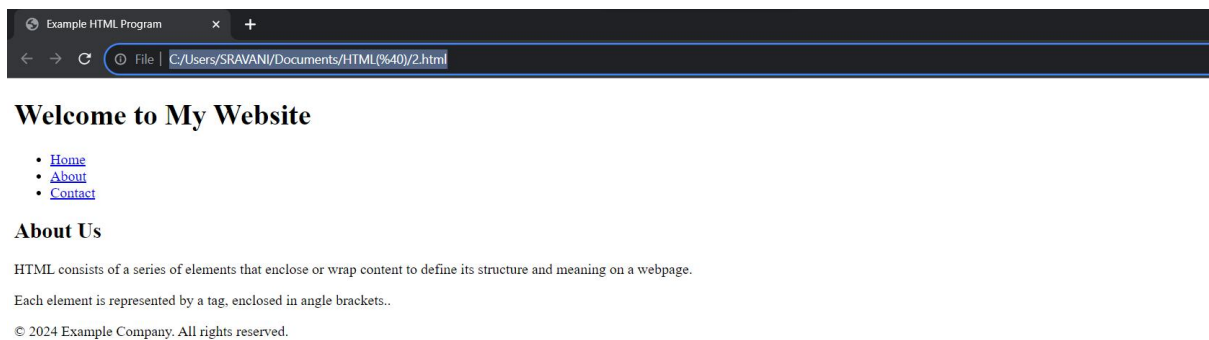
<p>© 2024 Example Company. All rights reserved. </p>

</footer>

</body>

</html>

Output



3. Design a web page demonstrating the usage of HTML Headers

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>HTML Headings Example</title>
```

```
</head>
```

```
<body>
```

```
<h1>This is a Heading Level 1</h1>
```

```
<p>This is some content under Heading 1.</p>
```

```
<h2>This is a Heading Level 2</h2>
```

```
<p>This is some content under Heading 2.</p>
```

```
<h3>This is a Heading Level 3</h3>
```

```
<p>This is some content under Heading 3.</p>
```

```
<h4>This is a Heading Level 4</h4>
```

```
<p>This is some content under Heading 4.</p>
```

<h5>This is a Heading Level 5</h5>

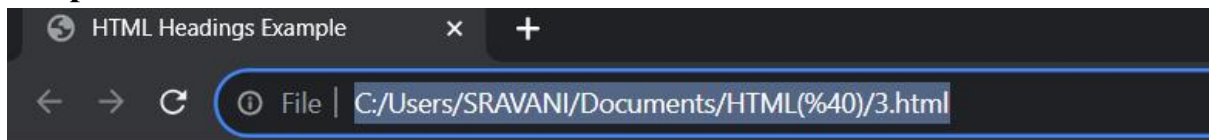
<p>This is some content under Heading 5.</p>

<h6>This is a Heading Level 6</h6>

<p>This is some content under Heading 6.</p>

</body>

Output



This is a Heading Level 1

This is some content under Heading 1.

This is a Heading Level 2

This is some content under Heading 2.

This is a Heading Level 3

This is some content under Heading 3.

This is a Heading Level 4

This is some content under Heading 4.

This is a Heading Level 5

This is some content under Heading 5.

This is a Heading Level 6

This is some content under Heading 6.

4.Design a web page demonstrating the usage of HTML paragraph tags

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>HTML Paragraphs Example</title>
```

```
</head>
```

```
<body>
```

```
    <p>This is a paragraph of text. It can contain multiple sentences and is  
    typically used to present textual content on a webpage.</p>
```

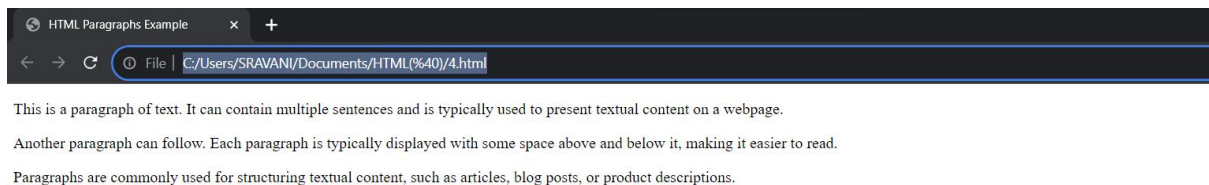
```
    <p>Another paragraph can follow. Each paragraph is typically displayed with  
    some space above and below it, making it easier to read.</p>
```

```
    <p>Paragraphs are commonly used for structuring textual content, such as  
    articles, blog posts, or product descriptions.</p>
```

```
</body>
```

```
</html>
```

Output:



5. Design a web page demonstrating the usage of HTML lists

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>HTML Lists Example</title>
```

```
</head>
```

```
<body>
```

```
<h2>Ordered List (Numbered)</h2>
```

```
<ol>
```

```
    <li>Item 1</li>
```

```
    <li>Item 2</li>
```

```
    <li>Item 3</li>
```

```
</ol>
```

```
<h2>Unordered List (Bullet Points)</h2>
```

```
<ul>
```

```
    <li>Apple</li>
```

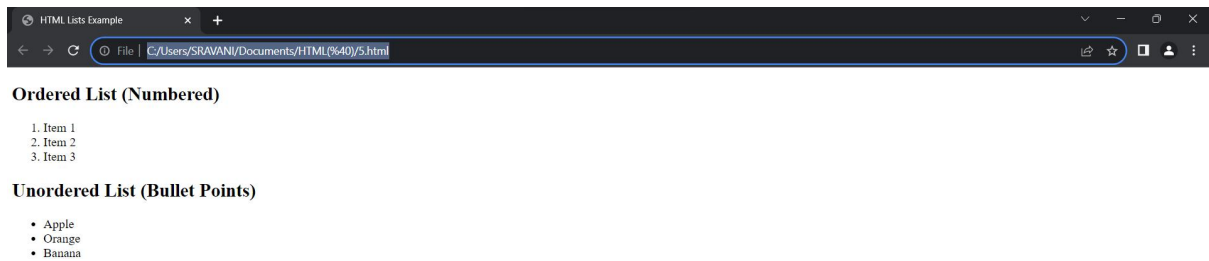
```
    <li>Orange</li>
```

```
    <li>Banana</li>
```


</body>

</html>

Output:



6. Design a web page demonstrating the usage of HTML links

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>HTML Links Example</title>

</head>

<body>

  <h2>Internal Links</h2>

  <p><a href="#section1">Jump to Section 1</a></p>

  <p><a href="#section2">Jump to Section 2</a></p>

  <h2>External Links</h2>

  <p><a href="https://www.externallink1.com">Visit Example Website</a></p>

  <p><a href="https://www.externallink2.com">Learn HTML on
Wikipedia</a></p>

  <h2>Email Links</h2>

  <p><a href="mailto:person@example.com">Send an Email to
Example</a></p>

  <h2>File Links</h2>

  <p><a href="documents/example.pdf">Download Example PDF</a></p>

  <h2>Linking to Images</h2>
```

```
<p><a href="images/example.jpg">View Example Image</a></p>
```

```
<h2>Linking to Video</h2>
```

```
<p><a href="videos/example.mp4">Watch Example Video</a></p>
```

```
<h2>Anchor Links</h2>
```

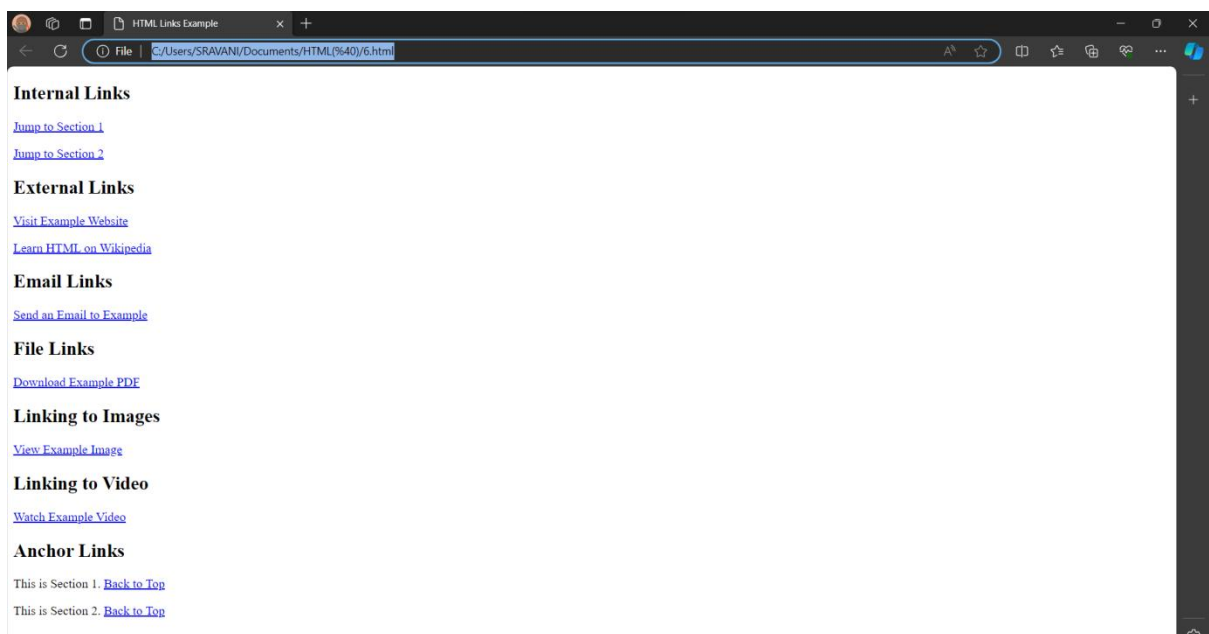
```
<p id="section1">This is Section 1. <a href="#">Back to Top</a></p>
```

```
<p id="section2">This is Section 2. <a href="#">Back to Top</a></p>
```

```
</body>
```

```
</html>
```

Output:



7. Design a web page demonstrating the usage of HTML table tags

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>HTML Table Example</title>
```

```
</head>
```

```
<body>
```

```
    <table border="1">
```

```
        <tr>
```

```
            <td>Row 1, Cell 1</td>
```

```
            <td>Row 1, Cell 2</td>
```

```
            <td>Row 1, Cell 3</td>
```

```
        </tr>
```

```
        <tr>
```

```
            <td>Row 2, Cell 1</td>
```

```
            <td>Row 2, Cell 2</td>
```

```
            <td>Row 2, Cell 3</td>
```

```
        </tr>
```

```
    </table>
```

<td>Row 3, Cell 1</td>

<td>Row 3, Cell 2</td>

<td>Row 3, Cell 3</td>

</tr>

</table>

</body>

</html>

Output:



The screenshot shows a web browser window with the title "HTML Table Example". The address bar displays the file path "C:/Users/SRAVANI/Documents/HTML(%40)/7.html". The browser content area shows a table with 3 rows and 3 columns. The cells contain the text "Row 1, Cell 1", "Row 1, Cell 2", "Row 1, Cell 3" in the first row, "Row 2, Cell 1", "Row 2, Cell 2", "Row 2, Cell 3" in the second row, and "Row 3, Cell 1", "Row 3, Cell 2", "Row 3, Cell 3" in the third row.

Row 1, Cell 1	Row 1, Cell 2	Row 1, Cell 3
Row 2, Cell 1	Row 2, Cell 2	Row 2, Cell 3
Row 3, Cell 1	Row 3, Cell 2	Row 3, Cell 3

8. Design a web page demonstrating the usage of HTML id attribute

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>HTML ID Example</title>

  <style>

    #unique-id {

      color: palevioletred;

    }

  </style>

</head>

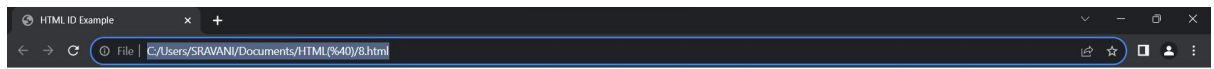
<body>

  <p id="unique-id">This paragraph has a unique ID.</p>

</body>

</html>
```

Output:



This paragraph has a unique ID.

9. Design a web page demonstrating the usage of HTML Class attribute

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>HTML Class Example</title>

    <style>

        .highlight {

            background-color:green ;

        }

    </style>

</head>

<body>

    <p class="highlight">This paragraph is highlighted. </p>

    <p>This paragraph is not highlighted. </p>

    <p class="highlight">Another highlighted paragraph. </p>

</body>

</html>
```

Output:



10. Design a web page demonstrating the usage of image tag in HTML

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>HTML Image Example</title>
```

```
</head>
```

```
<body>
```

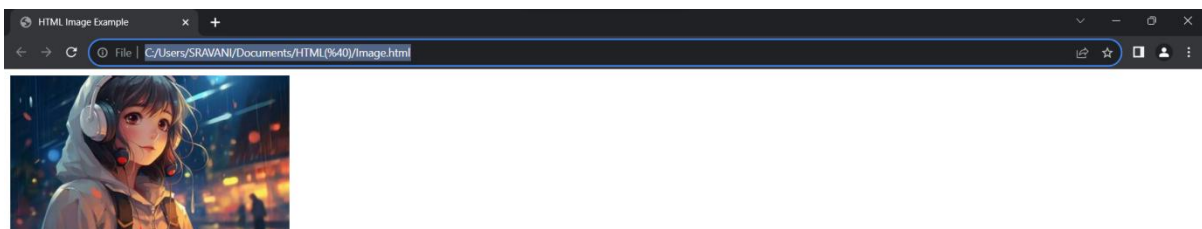
```

```

```
</body>
```

```
</html>
```

Output:



11. Design a web page demonstrating the usage of both HTML IDs & Classes stylings

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>HTML IDs & Classes Example</title>

    <style>

        /* Styling for elements with ID */

        #unique-id {

            color: violet;

        }

        /* Styling for elements with class */

        .highlight {

            background-color: orchid;

        }

    </style>

</head>

<body>
```


<!-- Element with ID -->

<p id="unique-id">This paragraph has a unique ID.</p>

<!-- Elements with class -->

<p class="highlight">This paragraph is highlighted.</p>

<p>This paragraph is not highlighted.</p>

<p class="highlight">Another highlighted paragraph.</p>

</body>

</html>

Output:



12. Design a web page demonstrating the usage of simple registration form

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>Registration Form</title>
```

```
</head>
```

```
<body>
```

```
    <h2>User Registration</h2>
```

```
    <form action="/submit_form" method="post">
```

```
        <label for="username">Username:</label><br>
```

```
        <input type="text" id="username" name="username"><br>
```

```
        <label for="email">Email:</label><br>
```

```
        <input type="email" id="email" name="email"><br>
```

```
        <label for="password">Password:</label><br>
```

```
        <input type="password" id="password" name="password"><br>
```

```
<label for="confirm_password">Confirm Password:</label><br>

<input type="password" id="confirm_password"
name="confirm_password"><br>

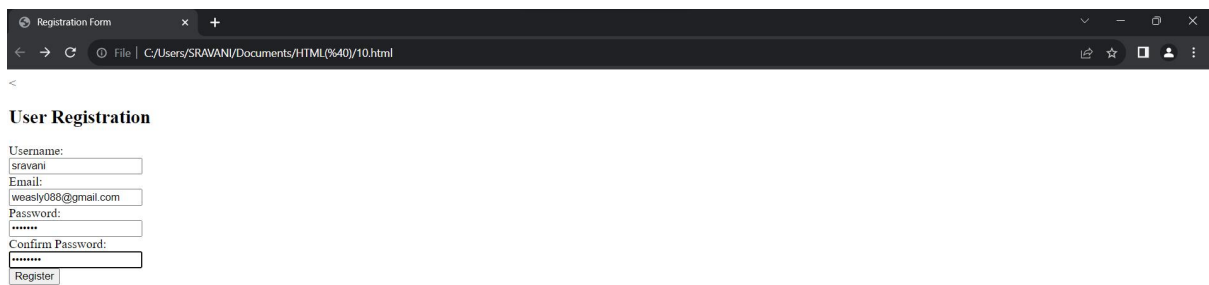
<input type="submit" value="Register">

</form>

</body>

</html>
```

Output



The screenshot shows a web browser window with the title 'Registration Form'. The address bar displays the file path 'C:/Users/SRAVANI/Documents/HTML(%40)/10.html'. The page content is titled 'User Registration' and contains a form with the following fields and controls:

- Username:** A text input field containing the value 'sravani'.
- Email:** A text input field containing the value 'weesly088@gmail.com'.
- Password:** A password input field with masked characters '*****'.
- Confirm Password:** A password input field with masked characters '*****'.
- Register:** A submit button located below the confirm password field.

13. Design a web page demonstrating the usage of radio buttons and checkboxes

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>HTML Forms Part 2: Form Elements</title>

</head>

<body>

    <form action="/submit_form" method="post">

        <label for="username">Username:</label>

        <input type="text" id="username" name="username"><br><br>

        <label for="password">Password:</label>

        <input type="password" id="password" name="password"><br><br>

        <label>Gender:</label><br>

        <input type="radio" id="male" name="gender" value="male">

        <label for="male">Male</label><br>

        <input type="radio" id="female" name="gender" value="female">
```

```
<label for="female">Female</label><br><br>
<label for="hobbies">Hobbies:</label><br>
<input type="checkbox" id="reading" name="hobbies" value="reading">
<label for="reading">Reading</label><br>
<input type="checkbox" id="sports" name="hobbies" value="sports">
<label for="sports">Sports</label><br>
<input type="checkbox" id="music" name="hobbies" value="music">
<label for="music">Music</label><br><br>
<label for="country">Country:</label>
<select id="country" name="country">
  <option value="usa">USA</option>
  <option value="uk">UK</option>
  <option value="canada">Canada</option>
</select><br><br>
<input type="submit" value="Submit">
</form>
</body>
</html>
```

Output:

HTML Forms Part 2: Form Elements

File | C:/Users/SRAVANI/Documents/HTML(%40)/check.html

Username:

Password:

Gender:

☐ Male

☒ Female

Hobbies:

☐ Reading

☐ Sports

☒ Music

Country:

14. Design a web page by implementing the GET and POST Methods in HTML Form Submission

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Simple Form Example</title>

</head>

<body>

<h2>Simple Form Submission</h2>

<form action="submit_get.php" method="get">

    <label for="username">Username:</label>

    <input type="text" id="username" name="username" required>

    <button type="submit">Submit (GET)</button>

</form>

<form action="submit_post.php" method="post">

    <label for="email">Email:</label>

    <input type="email" id="email" name="email" required>
```

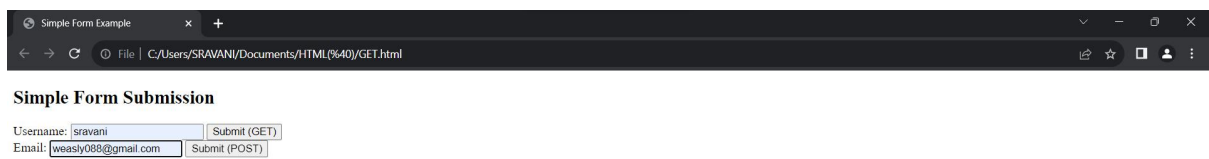
```
<button type="submit">Submit (POST)</button>
```

```
</form>
```

```
</body>
```

```
</html>
```

Output:



15. Design a webpage by implementing the Marquee tag and table rowspan and colspan

```
<DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width,initial-
scale=1.0">

    <title><Marquee> Tag and Rowspan/colspan Example</title>

  </head>

  <body>

    <marquee behaviour="scroll" direction="left" scroll amount="5">

      This is a scrolling marquee text:

    </marquee>

    <table border="1" align="center">

      <tr>

        <td rowspan="2">Row1,col1</td>

        <td colspan="2">Row1,col2 and Row1,col3</td>

      </tr>

      <tr>

        <td>Row2,col2</td>

        <td>Row2col3</td>
```

```

<tr>

<td>Row3,col3</td></td>>

</tr>

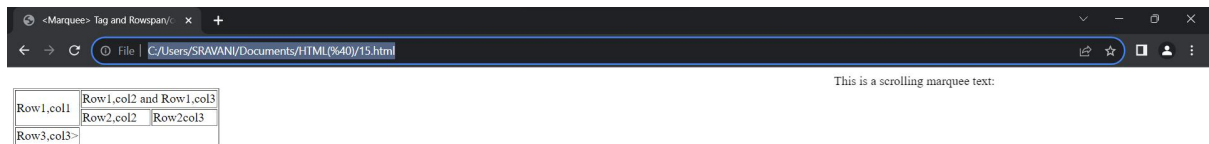
</table>

</body>

</html>

```

Output



16.Design a webpage by implementing the HTML video player

```
<!DOCTYPE html>
```

```
<html lang=="en">
```

```
<head>
```

```
<meta charset="UTF-8"/>
```

```
<meta name=""viewport"content="width=device-width,initial-scale=1.0"
```

```
<title>HTML videos</title>
```

```
</head>
```

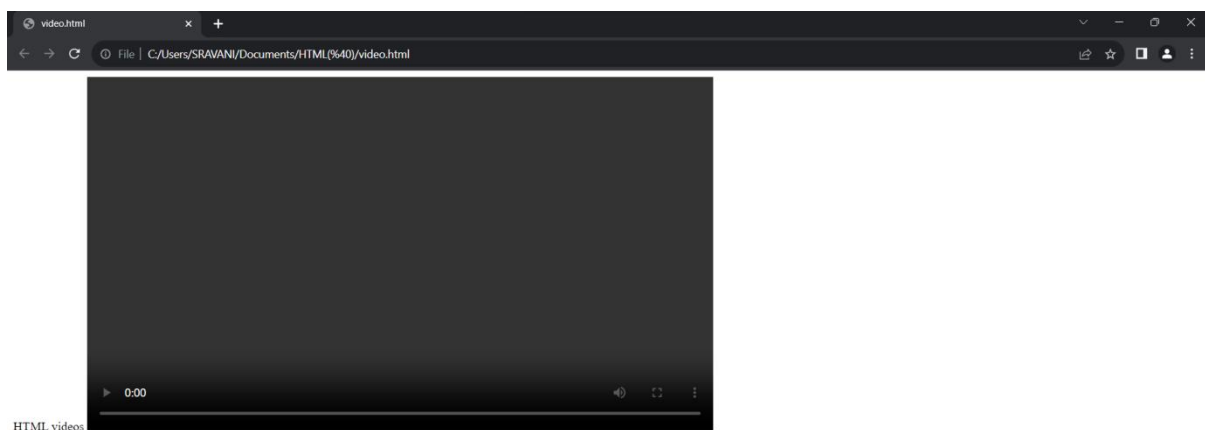
```
<body>
```

```
<video height="450"width="800"src="Aditya  
Promo.mp4"controls></video>
```

```
</body>
```

```
</html>
```

Output



17.Design a webpage by implementing the HTML Audio Player

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
  <head>
```

```
    <meta charset="UTF-8"/>
```

```
    <meta name="viewport" content="width=device-width,initial-scale=1.0"/>
```

```
    <title>HTML Audio</title>
```

```
  </head>
```

```
  <body>
```

```
    <audio src="Aditya-Promo.mp3"controls>
```

```
  </body>
```

```
</html>
```

Output



18. Write a program on HTML Semantics

```
<!doctype html>

<html lang="eng">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width,initial-scale=1.0">

<title>html semantic elements</title>

</head>

<body>

<header>

<nav>

<ul>

<li><a href="#"> Home</a></li>

<li><a href="#"> About</a></li>

<li><a href="#"> Service</a></li>

<li><a href="#"> Contact</a></li>

</ul>

</nav>

</header>

<h1> ANIMAL: </h1>

<section>

<h1> DOLPHINS </h1>
```

<p>

A dolphin is an aquatic mammal within the infraorder Cetacea.

</p>

</section>

<section>

<h2> LION </h2>

<p>

The lion (*Panthera leo*) is a large cat of the genus *Panthera*, native to Africa and India.

</section>

<articles>

<h2> CLIMATE CHANGE </h2>

<p>

Climate change has an increasingly large impact on the environment.

</p>

<details>

<summary>READ MORE</summary>

<p>

Climate change is happening globally, but the effects can differ locally.

</p>

<p>

key impacts of climate change include:

</p>

<p>

1 RISING TEMPERATURE:contribute to heat waves and other extreme weather events that can cause illness and death

redefine where plants and animals can survive. Many organisms cannot relocate or adapt quickly enough when warm or cool conditions change their environment. Other organisms, like some viruses, thrive in the altered habitats and can spread.

</p>

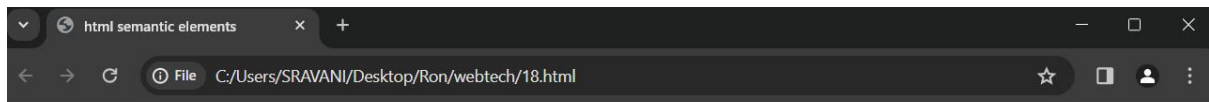
</details>

</article>

</body>

</html>

Output



- [Home](#)
- [About](#)
- [Service](#)
- [Contact](#)

ANIMAL:

DOLPHINS

A dolphin is an aquatic mammal within the infraorder Cetacea.

LION

The lion (*Panthera leo*) is a large cat of the genus *Panthera*, native to Africa and India.

CLIMATE CHANGE

Climate change has an increasingly large impact on the environment.

▼ READ MORE

Climate change is happening globally, but the effects can differ locally.

key impacts of climate change include:

1 RISING TEMPERATURE:contribute to heat waves and other extreme weather events that can cause illness and death redefine where plants and animals can survive. Many organisms cannot relocate or adapt quickly enough when warm or cool conditions change their environment. Other organisms, like some viruses, thrive in the altered habitats and can spread.

19. Write a program to create class time table

```
<!doctype html>

<html lan="en">

<head>

<title> time table </title>

</head>

<body bgcolor="white">

<header>

<h1> TIME TABLE </h1>

</header>

<table border="1">

<tr>

<td align="center" height="50" width="100"><br>

<b><day/period></b></br>

<td align="center" height="50" width="100">

<b>1<br>9:00-9:45</b>

</td>

<td align="center" height="50" width="100">

<b>2<br>9:45-10:30</b>

</td>

<td align="center" height="50" width="100">

<b>10:30-10:45</b>
```

</td>

<td align="center" height="50" width="100">

3
10:45-11:30

</td>

<td align="center" height="50" width="100">

4
11:30-12:15

</td>

<td align="center" height="50" width="100">

12:15-1:00

</td>

<td align="center" height="50" width="100">

5
1:00-1:45

</td>

<td align="center" height="50" width="100">

6
1:45-2:30

</td>

<td align="center" height="50" width="100">

7
2:30-3:15

</td>

<td align="center" height="50" width="100">

3:15-3:30

</td>

```

<td align="center" height="50" width="100">

<b>8<br>3:30-4:15</b>

</td>

</tr>

<tr>

<td align="center" height="50">

<b> MONDAY </b>

</td>

<td align="center" height="50">TEL</td>

<td align="center" height="50">HIN</td>

<td rowspan="6" align="center" height="50">

<h2>B<br>R<br>E<br>A<br>K<br></h2>

</td>

<td colspan="2" align="center" height="50">MAT</td>

<td rowspan="6" align="center" height="50">

<h2>L<br>U<br>N<br>C<br>H<br></h2>

</td>

<td align="center" height="50">ENG</td>

<td align="center" height="50">CS Lab</td>

<td align="center" height="50">C Language</td>

<td rowspan="6" align="center" height="50">

<h2>B<br>R<br>E<br>A<br>K<br></h2>

```

```

</td>

<td align="center" height="50">SOC</td>

</td>

</tr>

<tr>

<td align="center" height="50">

<b> TUESDAY </b>

</td>

<td align="center" height="50">Opearing System</td>

<td align="center" height="50">Java</td>

<td align="center" height="50">Python</td>

<td align="center" height="50">Digital Imaging</td>

<td align="center" height="50">Data Analytics Using R</td>

<td align="center" height="50">ENG</td>

<td align="center" height="50">CS Lab</td>

<td align="center" height="50">Data Structures</td>

</td>

</tr>

<td align="center" height="50">

<b> WEDNESDAY </b>

</td>

<td align="center" height="50">Opearing System</td>

```

```

<td align="center" height="50">Java</td>

<td align="center" height="50">Python</td>

<td align="center" height="50">Digital Imaging</td>

<td align="center" height="50">Data Analytics Using R</td>

<td align="center" height="50">ENG</td>

<td align="center" height="50">CS Lab</td>

<td align="center" height="50">Data Structures</td>

</td>

</tr>

<tr>

<td align="center" height="50">

<b> THUESDAY </b>

</td>

<td align="center" height="50">Opearing System</td>

<td align="center" height="50">Java</td>

<td align="center" height="50">Python</td>

<td align="center" height="50">Digital Imaging</td>

<td align="center" height="50">Data Analytics Using R</td>

<td align="center" height="50">ENG</td>

<td align="center" height="50">CS Lab</td>

<td align="center" height="50">Data Structures</td>

</td>

```

```

</tr>

<tr>

<td align="center" height="50">

<b> FRIDAY </b>

</td>

<td align="center" height="50">Operating System</td>

<td align="center" height="50">Java</td>

<td align="center" height="50">Python</td>

<td align="center" height="50">Digital Imaging</td>

<td align="center" height="50">Data Analytics Using R</td>

<td align="center" height="50">ENG</td>

<td align="center" height="50">CS Lab</td>

<td align="center" height="50">Data Structures</td>

</td>

</tr>

<tr>

<td align="center" height="50">

<b> SATURDAY </b>

</td>

<td align="center" height="50">Operating System</td>

<td align="center" height="50">Java</td>

<td align="center" height="50">Python</td>

```

```

<td align="center" height="50">Digital Imaging</td>

<td align="center" height="50">Data Analytics Using R</td>

<td align="center" height="50">ENG</td>

<td align="center" height="50">CS Lab</td>

<td align="center" height="50">Data Structures</td>

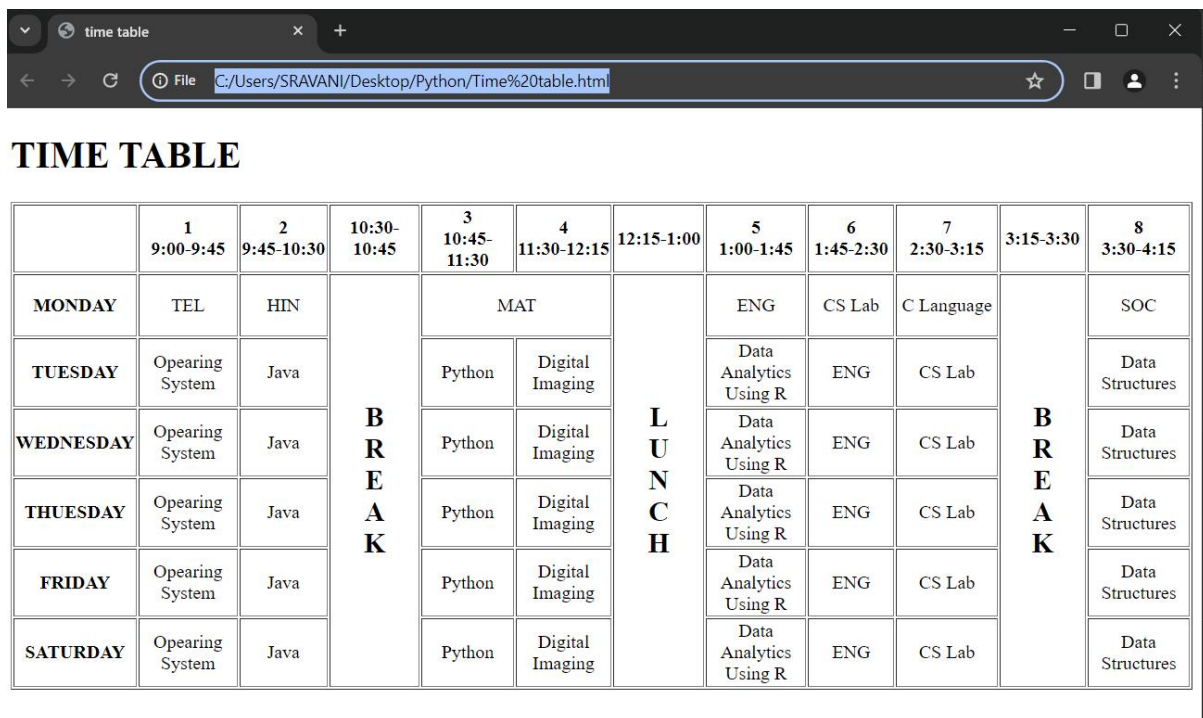
</td>

</tr>

</body>

```

Output



	1 9:00-9:45	2 9:45-10:30	10:30- 10:45	3 10:45- 11:30	4 11:30-12:15	12:15-1:00	5 1:00-1:45	6 1:45-2:30	7 2:30-3:15	3:15-3:30	8 3:30-4:15
MONDAY	TEL	HIN	B R E A K	MAT		L U N C H	ENG	CS Lab	C Language	B R E A K	SOC
TUESDAY	Opearing System	Java		Python	Digital Imaging		Data Analytics Using R	ENG	CS Lab		Data Structures
WEDNESDAY	Opearing System	Java		Python	Digital Imaging		Data Analytics Using R	ENG	CS Lab		Data Structures
THUESDAY	Opearing System	Java		Python	Digital Imaging		Data Analytics Using R	ENG	CS Lab		Data Structures
FRIDAY	Opearing System	Java		Python	Digital Imaging		Data Analytics Using R	ENG	CS Lab		Data Structures
SATURDAY	Opearing System	Java		Python	Digital Imaging		Data Analytics Using R	ENG	CS Lab		Data Structures

CSS

1. Design a web page exploring different types of CSS Application

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="CSS_intro.css" />

    <title>CSS Introduction</title>

  </head>

  <body>

    <!-- Inline CSS ->>> BAD BAD Practice -->

    <!-- Internal CSS ->>> used in small amount of code -->

    <!-- External CSS ->>> Best practice -->

    <!-- <h1 style="font-size: 100px">Some Heading</h1> -->

    <h1>Sravani</h1>

  </body>

</html>
```

CSS 1


```
h1 {  
    font-size: 120px;  
    Color:palevioletred;  
}
```

Output



Sravani

2.Design a web page styling with different Colour Values

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8" />

<meta name="viewport" content="width=device-width, initial-
scale=1.0" />

<link rel="stylesheet" href="CSS_colors.css" />

<title>CSS Colors</title>

</head>

<body>

<section>

<h1>Welcome to CSS</h1>

<p>CSS frameworks like Bootstrap, Foundation, and Bulma
    provide pre-written CSS and JavaScript components to streamline
    web development</p>

</section>

</body>

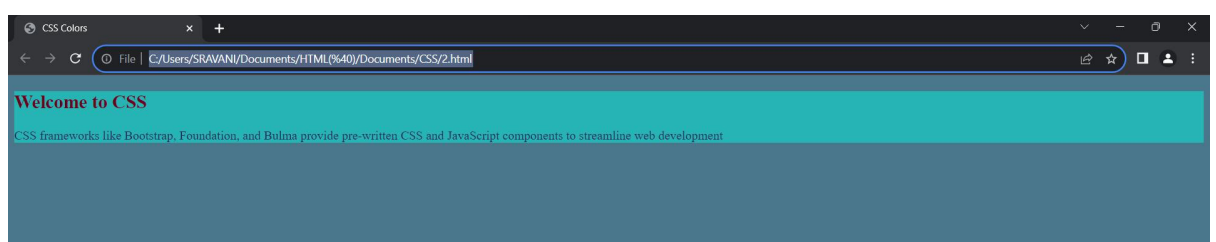
</html>
```

css 2

h1

```
body {  
  
    background-color: #4b778d;  
  
}  
  
section {  
  
    background-color: hsl(180, 64%, 43%);  
  
}  
  
h1 {  
  
    color: rgb(112, 0, 28);  
  
}  
  
p {  
  
    color: rgb(45, 45, 93);  
  
}
```

Output:



3.Design a web page styled with Simple CSS Selectors

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8" />

<meta name="viewport" content="width=device-width, initial-
scale=1.0" />

<link rel="stylesheet" href="3.css" />

<title>CSS Simple Selectors</title>

</head>

<body>

<section id="course">

<h1 class="title">CSS Bootcamp</h1>

<p class="para">You will learn CSS here</p>

</section>

</body>

</html>
```

Css 3

```
body {

    font-family: Arial, Helvetica, sans-serif;

}
```

```

#course {

    padding: 25px;

}

.title {

    font-size: 35px;

    color:blueviolet

}

* {

    margin: 0;

    padding: 0;

}

* {.title,

    .para {

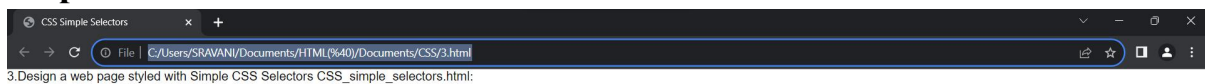
        margin: 10px;

        padding: 15px;

    }

```

output



CSS Bootcamp

You will learn CSS here

4.Design a web page demonstrating the usage of CSS background

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8" />
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
```

```
<link rel="stylesheet" href="4.css" />
```

```
<title>CSS Backgrounds</title>
```

```
</head>
```

```
<body>
```

```
<section>
```

```
<h1>Title 1</h1>
```

```
<p>
```

Lorem ipsum, dolor sit amet consectetur adipisicing elit. Voluptas

repellat officia recusandae omnis, necessitatibus quas quod?

Voluptatibus esse minus at perspiciatis, dolore, nulla, veniam voluptas blanditiis eaque doloribus asperiores. Nesciunt sit delectus magnam vitae atque eaque neque deserunt perferendis distinctio dolorem,

```
</p>
```

```
</section>
```

```
<section>
```

<h1>Title 2</h1>

<p>

Lorem ipsum, dolor sit amet consectetur adipisicing elit. Voluptas repellat officia recusandae omnis, necessitatibus quas quod?

Voluptatibus esse minus at perspiciatis, dolore, nulla, veniam voluptas

</p>

</section>

<section>

<h1>Title 3</h1>

<p>

Lorem ipsum, dolor sit amet consectetur adipisicing elit. Voluptas repellat officia recusandae omnis, necessitatibus quas quod?

</p>

</section>

<section>

<h1>Title 4</h1>

<p>

Lorem ipsum, dolor sit amet consectetur adipisicing elit. Voluptas repellat officia recusandae omnis, necessitatibus quas quod?

Voluptatibus esse minus at perspiciatis.

</p>

</section>

<section>

<h1>Title 5</h1>

<p>

Lorem ipsum, dolor sit amet consectetur adipisicing elit. Voluptas
repellat officia recusandae omnis, necessitatibus quas quod?

Voluptatibus esse minus.

</p>

</section>

<section>

<h1>Title 6</h1>

<p>

Lorem ipsum, dolor sit amet consectetur adipisicing elit. Voluptas
repellat officia recusandae omnis, necessitatibus quas quod?

Voluptatibus esse minus at perspiciatis, dolore, nulla, veniam.

</p>

</section>

<section>

<h1>Title 7</h1>

<p>

Lorem ipsum, dolor sit amet consectetur adipisicing elit. Voluptas

repellat officia recusandae omnis, necessitatibus quas quod?

lore accusamus beatae tenetur. Placeat, hic pariatursuscipit.

</p>

</section>

<section>

<h1>title 8</h1>

<p>

Consequatur, vel quis obcaecati labore doloribus unde id ipsum vitae

laborum, vero voluptate. Voluptates consectetur dicta repellendus

hic obcaecati modi cumque vero nemo impedit in odit eius nihil?

Quibusdam, deleniti?

Accusantium minus cupiditate odio enim, tempora excepturi nemo.

</p>

</section>

</body>

</html>

Css 4

body {

color: rgb(28, 147, 50);

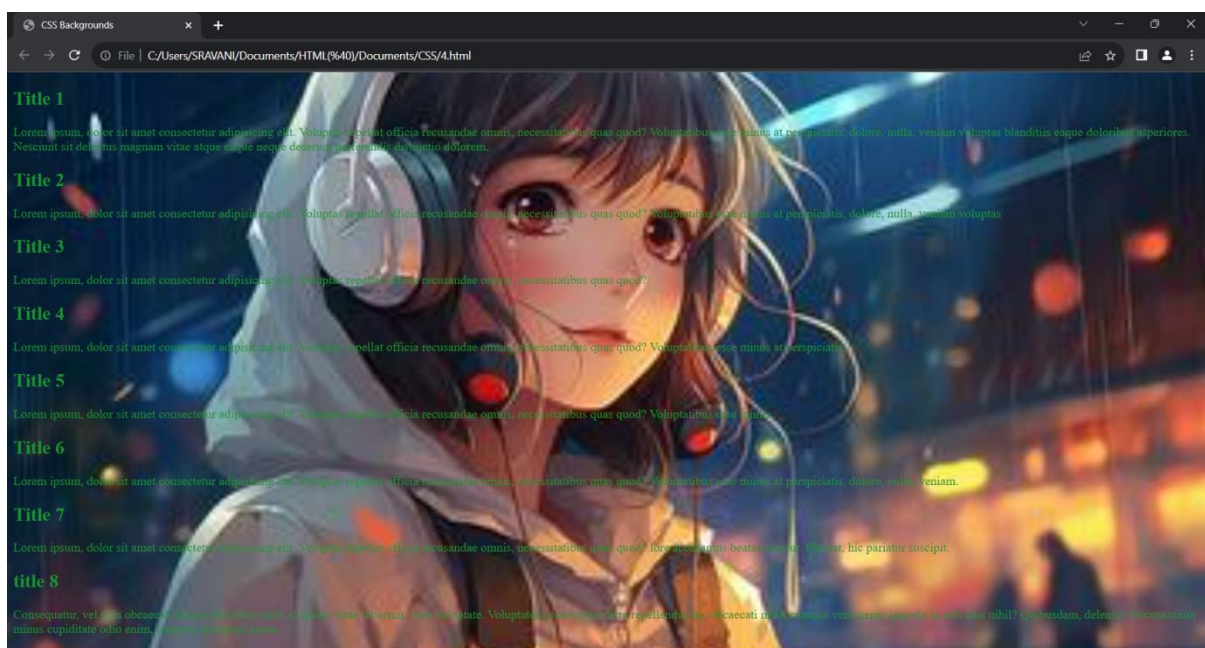
background-image: url("cartoon-lofi-young-manga-style-girl-while-listening-to-music-in-the-rain-ai-generative-photo.jpg");

```
background-position: center;
```

```
background-size: cover;
```

```
}
```

Output



5.Design a web page demonstrating the usage of CSS borders

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="5.css" />

    <title>CSS Borders</title>

  </head>

  <body>

    <section>

      <h1 class="title">CSS Borders</h1>

      <p class="para">In this page, you will learn how to set borders</p>

    </section>

  </body>

</html>
```

Css 5

```
section {

  border-style: solid;

  border-color:purple;

  border-width: 2px;

  padding: 10px;
```

```
} */
```

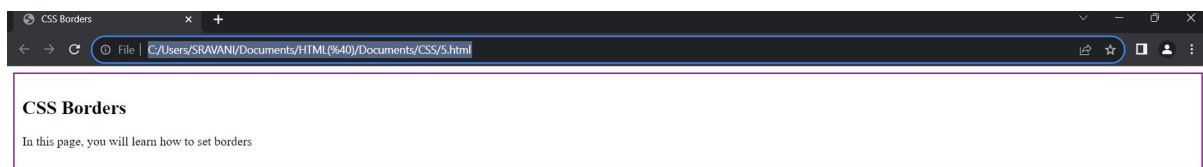
```
section {
```

```
    /* border: 10px dotted green; */
```

```
    border: indigo dashed 3px;
```

```
}
```

Output



6.Design a web page demonstrating the usage of CSS Margins

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="6.css" />

    <title>CSS Margins</title>

  </head>

  <body>

    <section>

      <h1 class="title">CSS Bootcamp</h1>

      <p class="para">You will learn how CSS borders styling works</p>

    </section>

  </body>

</html>
```

Css 6

```
/ section {

  margin-top: 15px;

  margin-right: 20px;
```

```
margin-bottom: 25px;
```

```
margin-left: 30px;
```

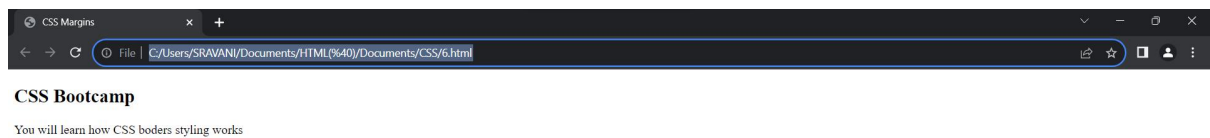
```
} */
```

```
section {
```

```
margin: 15px 20px 25px 30px;
```

```
}
```

Output



7.Design a web page demonstrating the usage of CSS Paddings

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="7.css" />

    <title>CSS Paddings</title>

  </head>

  <body>

    <section>

      <h1 class="title">CSS Bootcamp</h1>

      <p class="para">You will learn CSS from zero to advanced</p>

    </section>

  </body>

</html>
```

Css 7

```
/section {

  padding-top: 15px;
```

```
padding-right: 20px;

padding-bottom: 25px;

padding-left: 30px;

color:blue;

} */


section {

padding: 15px 20px 25px 30px;

}
```

Output



8.Design a web page demonstrating the usage of CSS Height & Width

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="8.css" />

    <title>CSS Height & Width</title>

  </head>

  <body>

    <section>

      <h1 class="title">CSS Bootcamp</h1>

      <p class="para">You will learn CSS.</p>

    </section>

  </body>

</html>
```

css 8

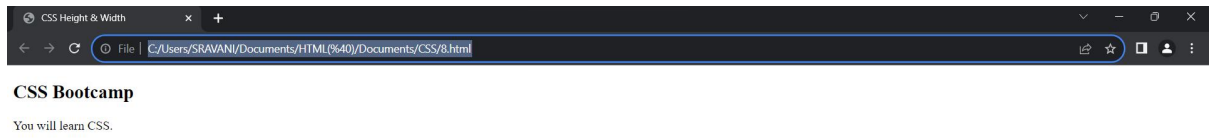
```
section {

  height: 200px;

  width: 600px;

}
```

Output



9.Design a web page demonstrating the usage of CSS Box mod

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="9.css" />

    <title>CSS Box Model</title>

  </head>

  <body>

    <section>

      <h1 class="title">CSS Bootcamp</h1>

      <p class="para">You will learn CSS from here. </p>

    </section>

    <article>

      <h1 class="title">CSS Bootcamp</h1>

      <p class="para">You will learn CSS. </p>

    </article>

  </body>

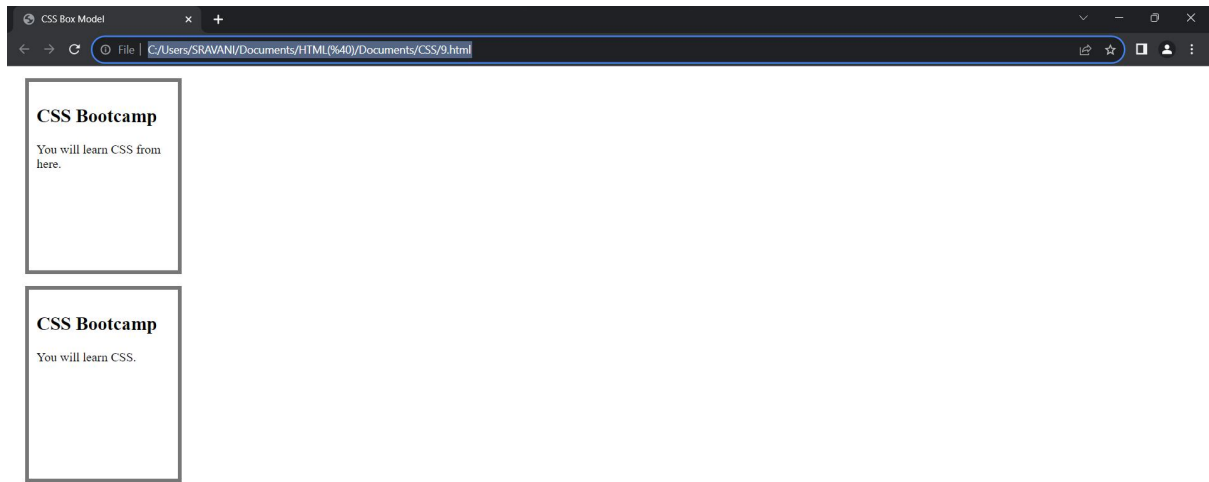
</html>
```

```
* {  
  
  /* box-sizing: content-box; */  
  
  box-sizing: border-box;  
  
}
```

```
section {  
  
  height: 250px;  
  
  width: 200px;  
  
  padding: 10px;  
  
  border: solid 5px #777;  
  
  margin: 15px;  
  
}
```

```
article {  
  
  height: 250px;  
  
  width: 200px;  
  
  padding: 10px;  
  
  margin: 15px;  
  
  border: solid 5px #777;  
  
}
```

Output



10.Design a web page demonstrating the usage of CSS text property.

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="10.css" />

    <title>CSS Text Properties</title>

  </head>

  <body>

    <section>

      <h1 class="title">CSS Bootcamp</h1>

      <p class="para">You will be learning CSS. </p>

    </section>

  </body>

</html>
```

Css 10

```
h1 {

  color:purple;

  text-transform: uppercase;

  text-align: center;
```

```
}
```

```
p {
```

```
text-align: center;
```

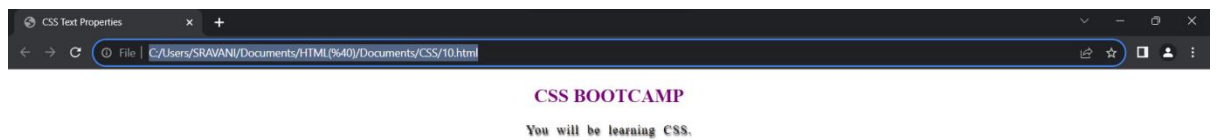
```
letter-spacing: 1px;
```

```
word-spacing: 5px;
```

```
text-shadow: 1px 2px 2px black;
```

```
}
```

Output



11.Design a web page demonstrating the usage of CSS Font properties.

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="11.css" />

    <title>CSS Font Properties</title>

  </head>

  <body>

    <section>

      <h1 class="title">CSS Bootcamp</h1>

      <p class="para">You will learn CSS here.</p>

    </section>

  </body>

</html>
```

css 11

```
section {

  font-family: Verdana, Geneva, Tahoma, sans-serif;

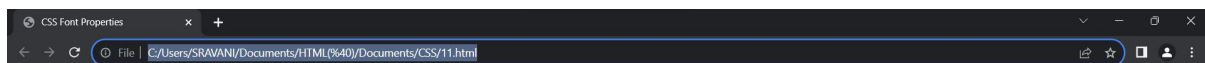
}
```



```
h1 {  
  
  color:blueviolet;  
  
  text-transform: uppercase;  
  
  text-align: center;  
  
  
  font-style: italic;  
  
  font-size: 50px;  
  
  font-weight: 900;  
  
}
```

```
p {  
  
  text-align: center;  
  
  letter-spacing: 2px;  
  
  word-spacing: 5px;  
  
}
```

Output



CSS BOOTCAMP

You will learn CSS here.

12.Design a web page demonstrating the usage of CSS Links

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="12.css" />

    <title>CSS Links</title>

  </head>

  <body>

    <section>

      <h1 class="title">CSS Bootcamp</h1>

      <p class="para">You will learn css.</p>

      <a href="#">Read More</a>

    </section>

  </body>

</html>
```

css 12

```
section {

  font-family: Arial, Helvetica, sans-serif;
```

```
display: flex;

flex-direction: column;

justify-content: center;

align-items: center;

}
```

```
h1 {

    color: brown;

    text-transform: uppercase;

    font-size: 50px;

}
```

```
p {

    letter-spacing: 2px;

    word-spacing: 5px;

    font-weight: 700;

}
```

```
a {

    text-decoration: none;

    background-color: #333;

    padding: 15px;
```

```
margin: 40px;
```

```
color:violet;
```

```
transition: all 0.2s ease-in-out;
```

```
}
```

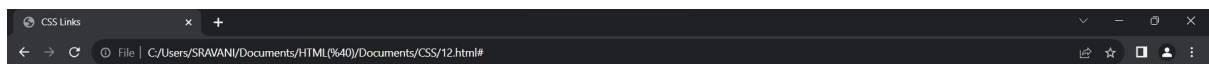
```
a:hover {
```

```
background-color: coral;
```

```
box-shadow: none;
```

```
}
```

Output



CSS BOOTCAMP

You will learn css.

Read More

13. Design a web page demonstrating the usage of CSS Combinators selectors

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="13.css" />

    <title>CSS Combinator Selectors</title>

  </head>

  <body>

    <article id="tech">

      <h1>Green Technology</h1>

      <p>

        Lorem ipsum dolor sit amet consectetur adipiscing elit. Accusantium

        nemo vel rem similique possimus! Porro, sint! Placeat, obcaecati iusto

        temporibus aspernatur accusantium sit est repellendus architecto neque

        fugit aliquid laudantium.

      </p>

      <a href="#">Read More</a>
```

```
<div>
```

```
<p>Hey There</p>
```

```
</div>
```

```
<a href="#">Read More Here</a>
```

```
<p>
```

Lorem ipsum dolor sit amet consectetur adipisicing elit. Dicta modi eum, cupiditate vitae earum commodi quas accusantium doloremque iusto sint nam quis possimus, consectetur sunt beatae nisi sed quibusdam quod.

```
</p>
```

```
</article>
```

```
</body>
```

```
</html>
```

css 13

```
#tech h1 {
```

```
font-size: 55px;
```

```
}
```

```
article > p {
```

```
background-color: lightsteelblue;

padding: 15px;

}
```

```
p ~ a {

text-decoration: none;

background-color: lightslategray;

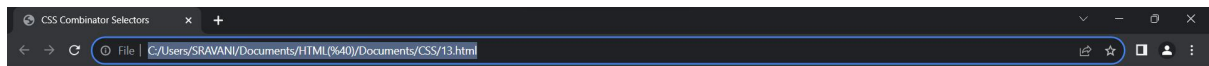
padding: 25px;


display: inline-block;

margin: 30px;

}
```

Output



Green Technology

Lorem ipsum dolor sit amet consectetur adipisicing elit. Accusantium nemo vel rem similique possimus! Porro, sint! Placeat, obcaecati iusto temporibus aspernatur accusantium sit est repellendus architecto neque fugit aliquid laudantium.

[Read More](#)

Hey There

[Read More Here](#)

Lorem ipsum dolor sit amet consectetur adipisicing elit. Dicta modi eum, cupiditate vitae earum commodi quas accusantium doloremque iusto sint nam quis possimus, consectetur sunt beatae nisi sed quibusdam quod.

14. Design a web page demonstrating the usage of CSS Pseudo classes

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8" />
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
```

```
<link rel="stylesheet" href="14.css" />
```

```
<title>CSS Pseudo Class Selectors</title>
```

```
</head>
```

```
<body>
```

```
<article id="animal-planet">
```

```
<h1>Polar Bears</h1>
```

```
<p class="para">
```

Lorem ipsum dolor sit amet consectetur adipisicing elit. Ex, corporis!

Reiciendis, earum doloremque libero quisquam aliquam omnis quis
quidem

tempora ducimus, commodi vel officia iusto, ex voluptatum

```
</p>
```

```
<a href="#">Read More</a>
```

<div>

<p>Hey there</p>

</div>

Read More Here

<p class="para main-para">

Lorem ipsum, dolor sit amet consectetur adipisicing elit. Ratione labore magni fugiat at recusandae voluptatem, corporis dolores veniam, aperiam quaerat quisquam, atque dolor.

</p>

Read More Here

<button>Click Me!!!</button>

</article>

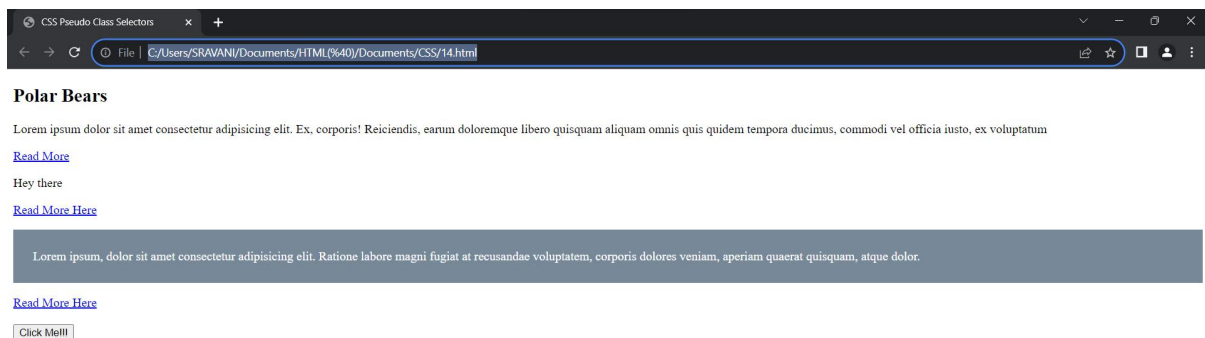
</body>

</html>

css 14

```
.main-para {  
  
    background-color: lightslategray;  
  
    padding: 25px;  
  
    color: white;  
  
}
```

Output



15.Design a web page demonstrating the usage of CSS opacity property

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="15.css" />

    <title>CSS Opacity</title>

  </head>

  <body>

    <p>

      Lorem ipsum dolor sit amet consectetur adipisicing elit. Quibusdam

      cupiditate quasi repellat nobis nihil, illo, aspernatur dolorum quis

      veniam consectetur accusamus? Aspernatur maiores illo odio. Labore nemo

      eos modi non?

    </p>

  </body>

</html>
```

css 15

```
body {
```

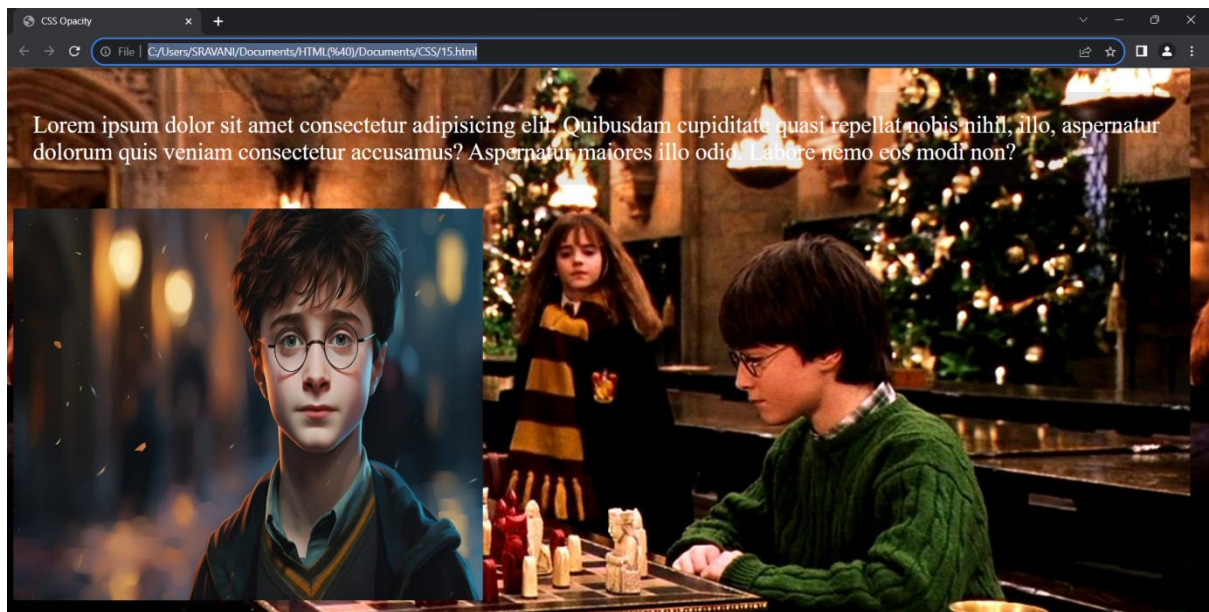
```
background-image: url("cartoon-lofi-young-manga-style-girl-while-  
listening-to-music-in-the-rain-ai-generative-photo.jpg");  
  
}
```

```
img {  
  
height: 500px;  
  
width: 600px;  
  
object-fit: calc("");  
  
}
```

```
img:hover {  
  
opacity: 0.2;  
  
}
```

```
p {  
  
color: #ffffff;  
  
padding: 25px;  
  
font-size: 30px;  
  
background-color: #4d1919a;  
  
}
```

Output



16.Design a web page demonstrating the usage of CSS Attribute Selectors

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta http-equiv="X-UA-Compatible" content="IE=edge" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="16.css" />

    <title>CSS Attribute Selectors</title>

  </head>

  <body>

    <article id="animal-planet">

      <h1 title="heading-one">Primates</h1>

      <p>

        Lorem ipsum dolor sit amet consectetur adipisicing elit. Ex, corporis!

        Reiciendis, earum doloremque libero quisquam aliquam omnis quis
        quidem

        tempora ducimus, commodi vel officia iusto.

      </p>

      <a href="#">Read More</a>
```

<div>

<p>Hey there</p>

</div>

Read More Here

<p>

Lorem ipsum, dolor sit amet consectetur adipisicing elit. Ratione labore magni fugiat at recusandae voluptatem, corporis dolores veniam, aperiam quaerat quisquam, atque dolor.

</p>

</article>

<article>

<h2>Birds</h2>

<p>

Lorem, ipsum dolor sit amet consectetur adipisicing elit. Culpa, cumque eveniet maiores adipisci doloribus dolorum.

</p>

</article>

</body>


```
</html>
```

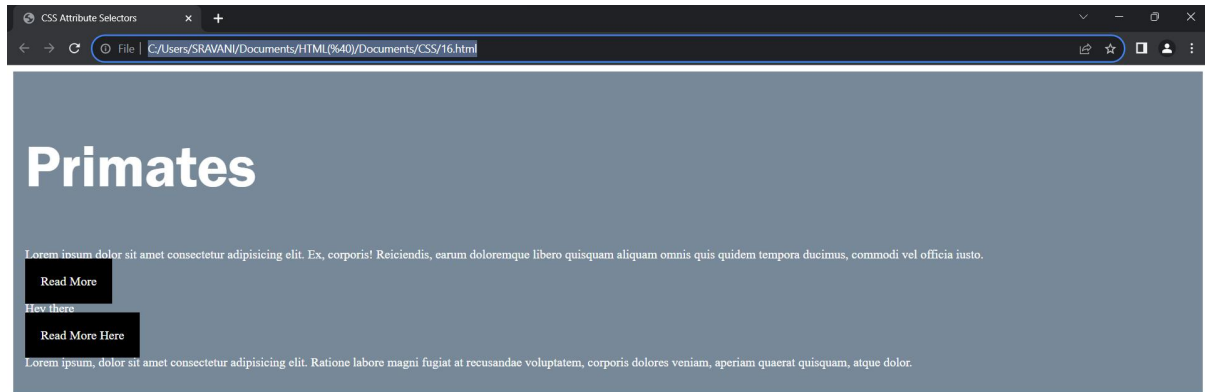
css 16

```
article[id="animal-planet"] {  
    background-color: lightslategray;  
    padding: 15px;  
    color: white;  
}
```

```
h1[title="heading-one"] {  
    font-size: 75px;  
    font-family: "Franklin Gothic Medium", "Arial Narrow", Arial, sans-serif;  
}
```

```
a[href="#"] {  
    text-decoration: none;  
    background-color: black;  
    color: white;  
    padding: 20px;  
}
```

Output



Birds

Lorem, ipsum dolor sit amet consectetur adipisicing elit. Culpa, cumque eveniet maiores adipisci doloribus dolorum.

17.Design a web page demonstrating the usage of CSS specificity

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-
scale=1.0" />

    <link rel="stylesheet" href="17.css" />

    <title>CSS Specificity</title>

  </head>

  <body>

    <section >

      <h1 id="title">CSS Bootcamp</h1>

      <p class="para">

        A comprehensive guide to learning one of the most used web

        languages of all time.

      </p>

      <a href="#">Read More</a>

    </section>

  </body>

</html>
```

css 17

```
section {  
    background-color: purple;  
}
```

```
#title {  
    font-size: 55px;  
}
```

```
h1 {  
    font-size: 40px;  
}
```

```
.para {  
    color: white;  
}
```

```
p {  
    color: blue;  
}
```

```
a {  
  
    font-size: 35px;  
  
}
```

```
h1#title {  
  
    font-size: 150px;  
  
}
```

```
#title {  
  
    font-size: 55px;  
  
}
```

Output



18.Design a web page demonstrating the usage of CSS Border Radius

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="18.css" />

    <title>CSS Border Radius</title>

  </head>

  <body>

    <section></section>

    <section></section>

  </body>

</html>
```

css 18

```
body {

  margin: 0;

  padding: 0;

  height: 100vh;

  display: flex;

  align-items: center;
```

```
}
```

```
section {  
  
    height: 350px;  
  
    width: 350px;  
  
    background-color: rgb(123, 53, 177);  
  
    margin: auto;  
  
  
    border-radius: 50%;  
}
```

Output



19.Design a web page demonstrating the usage of CSS Box shadow

```
!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="19.css" />

    <title>CSS Box Shadow</title>

  </head>

  <body>

    <section>

      <h1>CSS Bootcamp</h1>

      <p>

        A comprehensive guide to learning one of the most used web programming

        languages of all time

      </p>

    </section>

  </body>

</html>
```

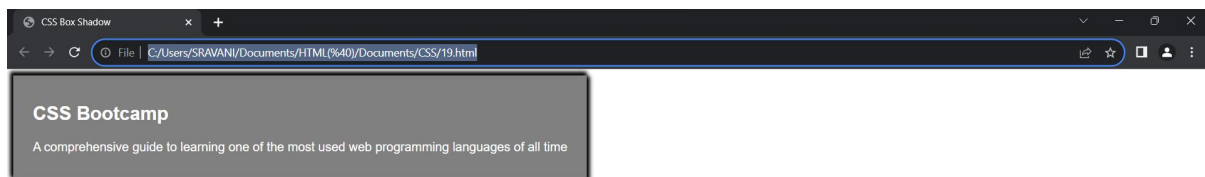
css 19

```
section {
```



```
padding: 15px 25px;  
  
color: #ffffff;  
  
font-family: Arial, Helvetica, sans-serif;  
  
background-color: rgb(128, 128, 128);  
  
display: inline-block;  
  
box-shadow: 2px 3px 5px 2px black, -2px -3px 5px 2px black;  
  
}
```

Output



20. Design a web page demonstrating the usage of CSS Gradients

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="20.css" />

    <title>CSS Gradients</title>

  </head>

  <body>

    <section>

      <h2>Chadwick Boseman</h2>

      <p>

        Lorem ipsum, dolor sit amet consectetur adipisicing elit. Sequi,

        possimus iusto? Sunt non exercitationem, voluptates corrupti odit,

        libero voluptatibus iste adipisci distinctio.

      </p>

    </section>

  </body>

</html>
```

```

    section {

padding: 25px;

color: white;

height: 500px;


/* Linear */

background-image: linear-gradient(#fff5ab, #0396ff);

background-image: linear-gradient(to right, #a143ff, #9708cc);

background-image: linear-gradient(to top, #92e2ff, #002661);

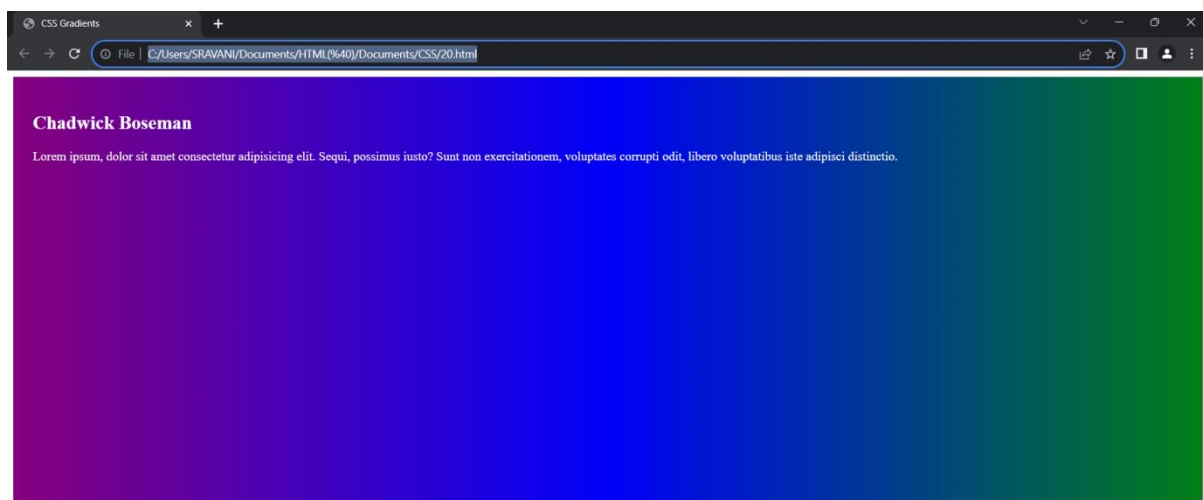
background-image: linear-gradient(to bottom, #ff92c3, #002661);

background-image: linear-gradient(to left, hsl(131, 100%, 25%), #0000f9,
rgb(134, 0, 125) );

}

```

Output



21.Design a web page demonstrating the usage of CSS transforms

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="21.css" />

    <title>CSS Transforms</title>

  </head>

  <body>

    <!-- div.block -->

    <div class="block">Here we go</div>

  </body>

</html>
```

css 21

```
.block {

  margin: 20px;

  height: 200px;

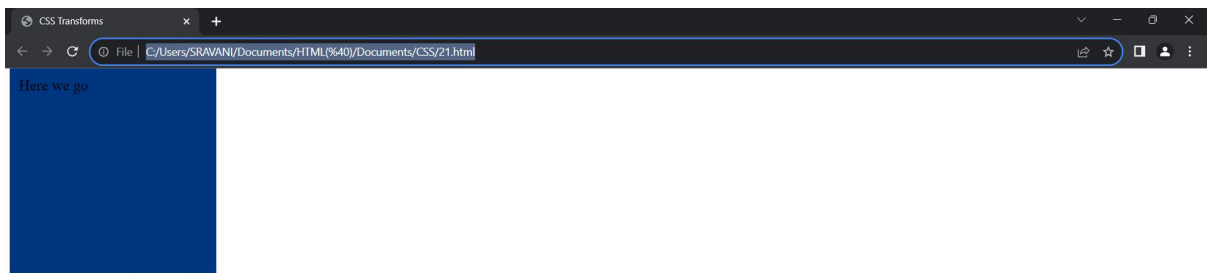
  width: 200px;

  padding: 10px;

  background-color: rgb(0, 53, 128);
```

```
transform: scale(1.2);  
  
}
```

Output



22.Design a web page demonstrating the usage of CSS transitions

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="22.css" />

    <title>CSS Transitions</title>

  </head>

  <body>

    <!-- div#block -->

    <div id="block"></div>

  </body>

</html>
```

css22

```
body {

  margin: 0;

  height: 100vh;

  display: grid;

  place-items: center;
```

```
}  
  
#block {  
  
    height: 250px;  
  
    width: 250px;  
  
    background-color: rgb(0, 255, 213);  
  
  
    transition: all 1s ease-in-out;  
  
}  
  
#block:hover {  
  
    transform: rotate(1turn);  
  
    border-radius: 50%;  
  
    background-color: black;  
  
}
```

Output



23.Design a web page demonstrating the usage of CSS animations

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="23.css" />

    <title>CSS Animations</title>

  </head>

  <body>

    <div class="block"></div>

  </body>

</html>
```

css 23

```
.block {

  height: 100px;

  width: 100px;

  background-color: rgb(252, 0, 38);

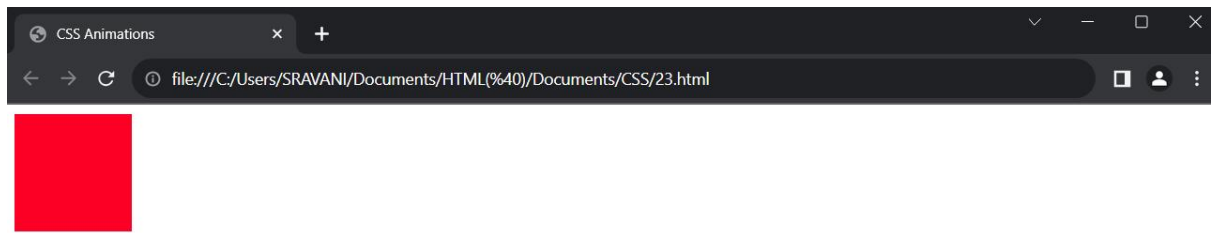
  animation-name: my-first-anime;

  animation-duration: 10s;

}
```

```
@keyframes my-first-anime {  
  0% {  
    transform: translate(0);  
  }  
  25% {  
    transform: translate(250px, 0);  
  }  
  50% {  
    transform: translate(250px, 150px);  
  }  
  75% {  
    transform: translate(0, 150px);  
  }  
  100% {  
    transform: translate(0);  
  }  
}
```

Output



24.Design a web page demonstrating the usage of CSS filters

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="24.css" />

    <title>CSS Filters</title>

  </head>

  <body>

  </body>

</html>
```

css 24

```
img {

  height: 600px;

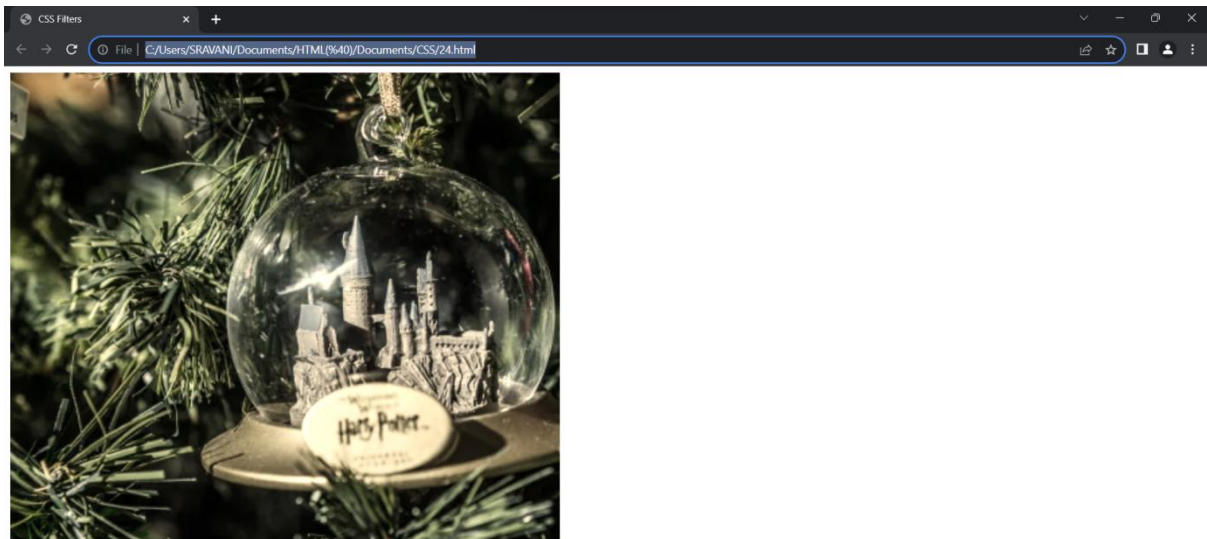
  width: 700px;

  object-fit: cover;

  filter: blur(10px);
```

```
filter: brightness(5%);  
  
filter: grayscale(100%);  
  
filter: contrast(10%);  
  
filter: opacity(0.8);  
  
filter: saturate(150%);  
  
filter: sepia(50%);  
  
}
```

Output



25.Design a web page demonstrating the usage of CSS clip path

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="25.css" />

    <title>CSS Clip Path</title>

  </head>

  <body>

  </body>

</html>
```

css 25

```
img {

  height: 500px;

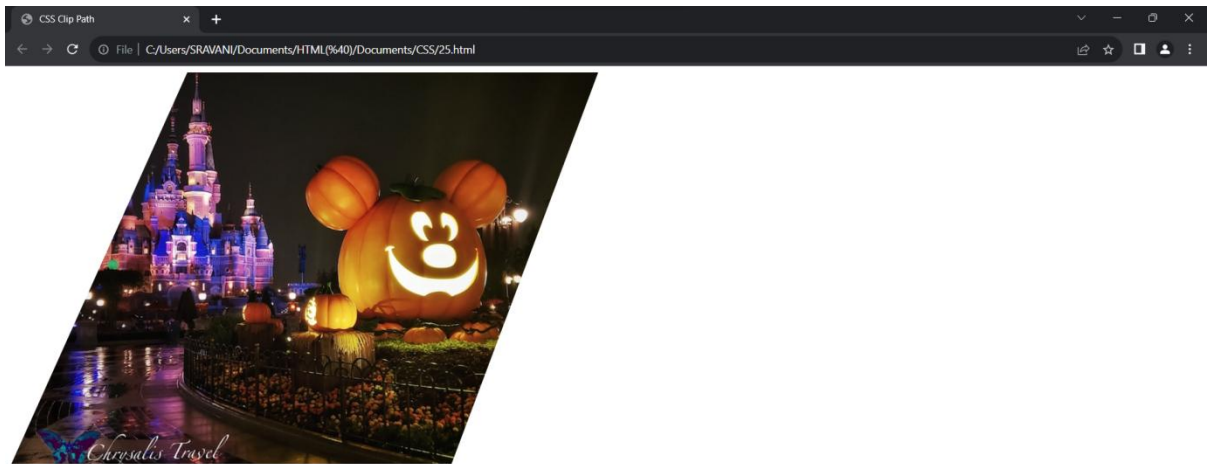
  width: 750px;

  object-fit: cover;

  clip-path: polygon(30% 0, 75% 0, 100% 0, 75% 100%, 0 100%, 0 100%);

}
```

Output

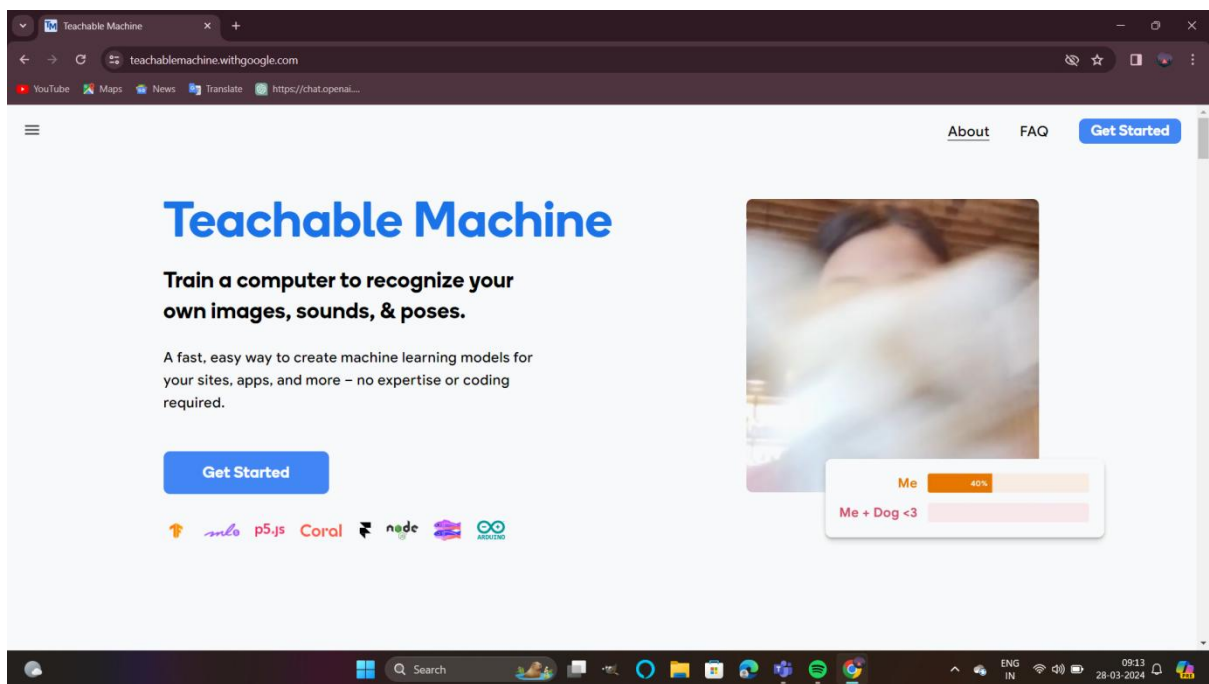


TEACHABLE MACHINE

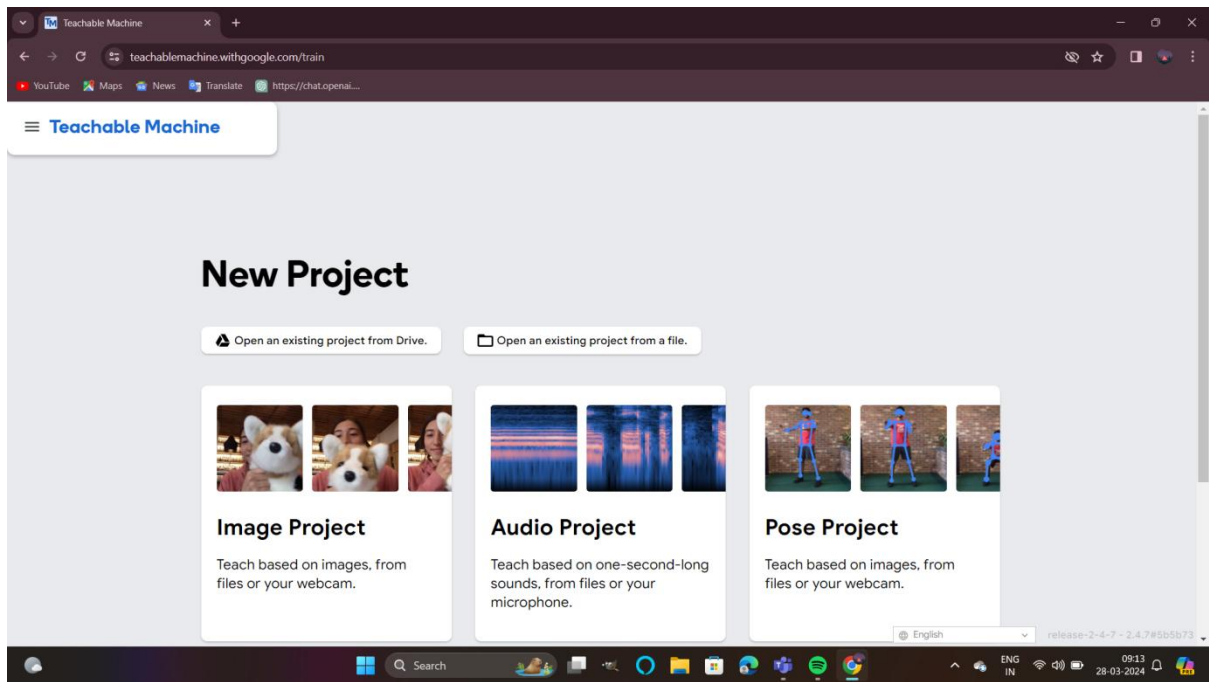
Title: Machine Learning Model predicting the output for the given user input

Use the link below:

<https://teachablemachine.withgoogle.com/>

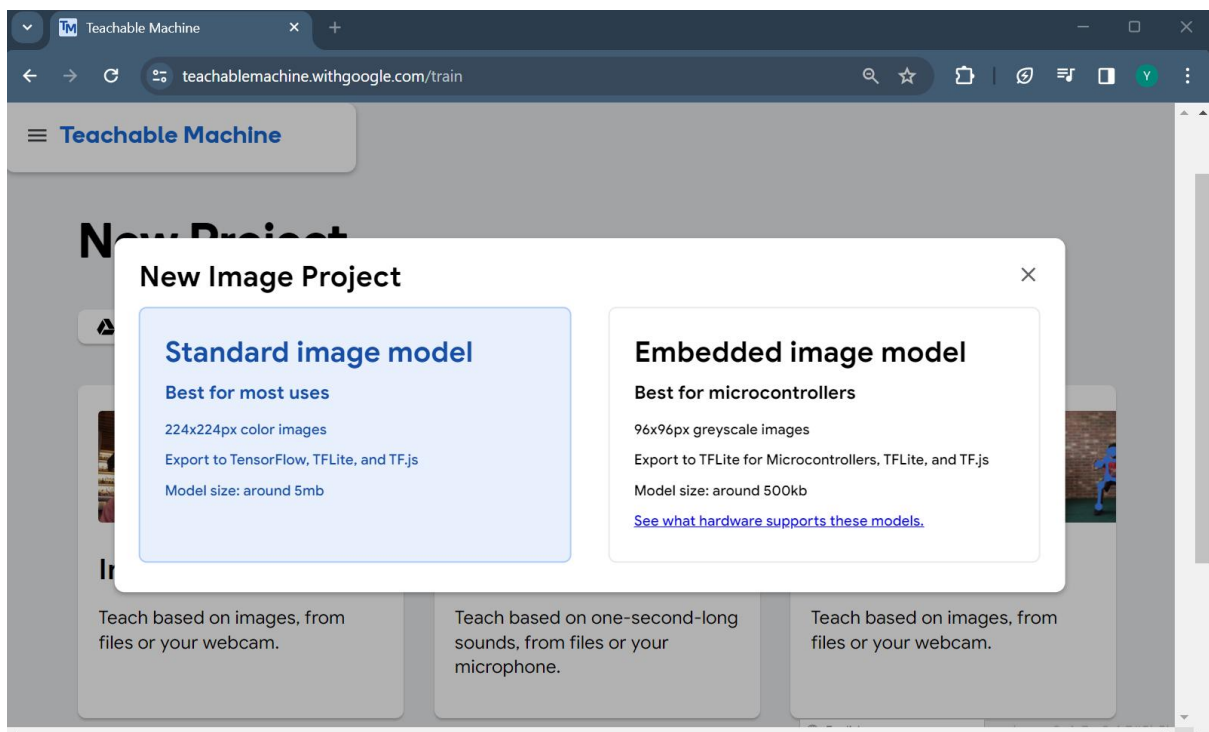


Click on get started

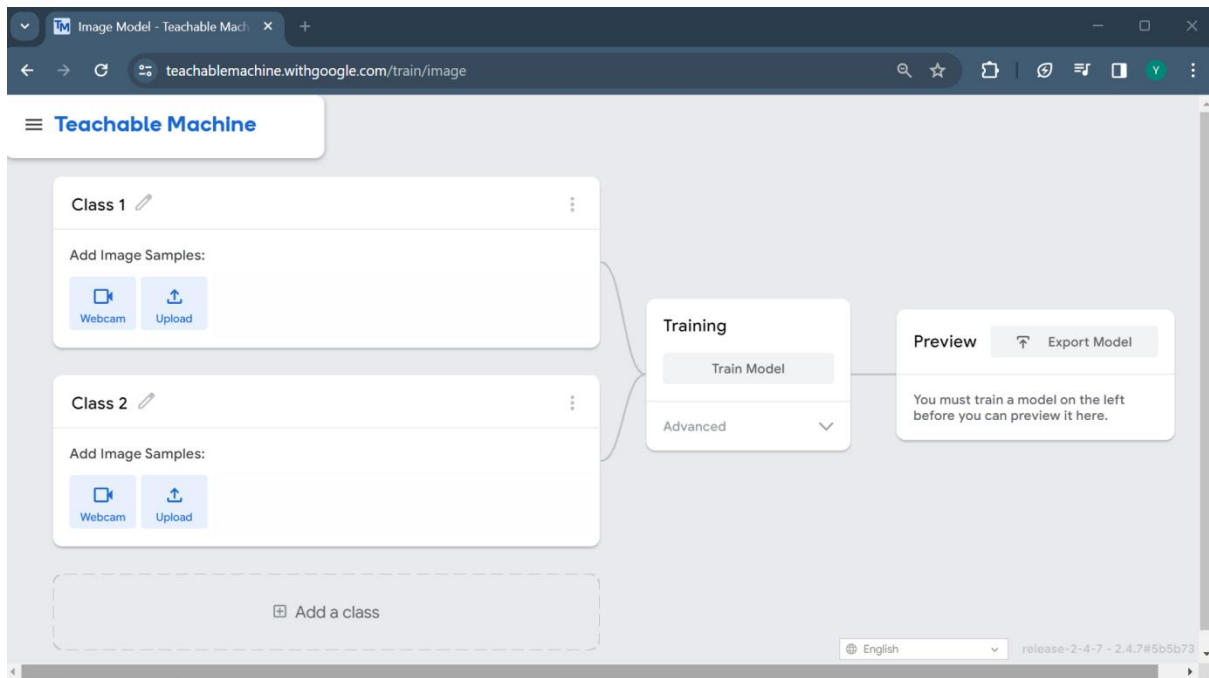


You can create projects and can play with these models

1. Image Project model:

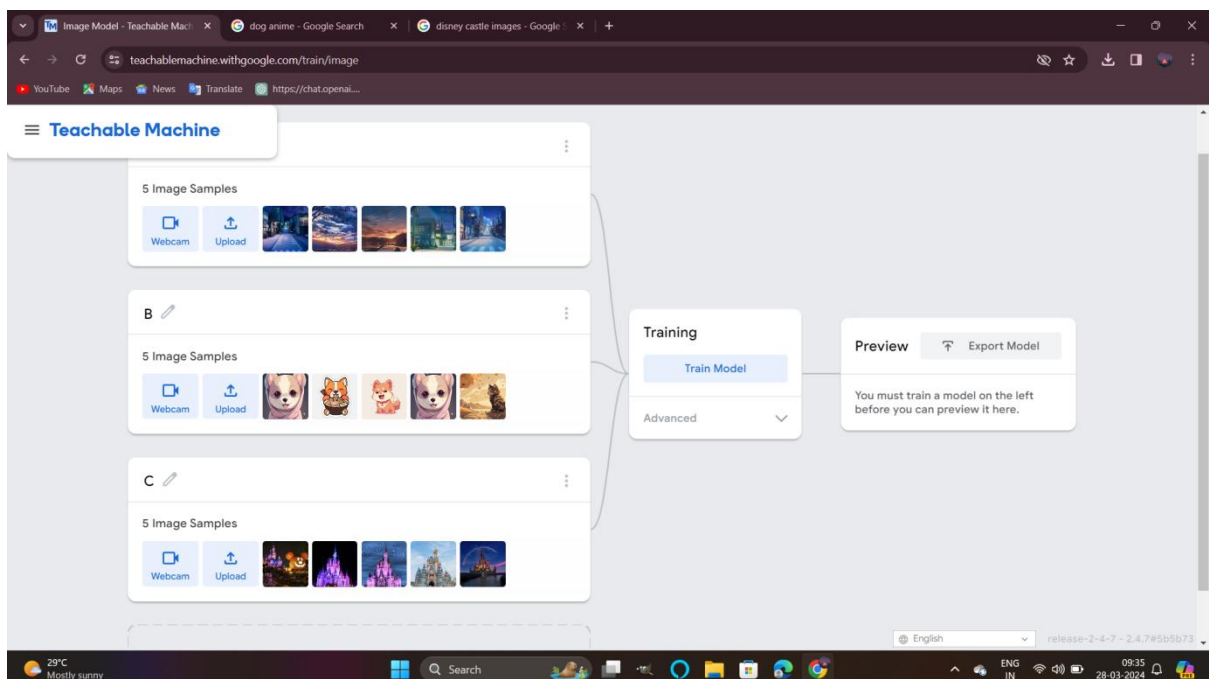


Choose **Standard image model** Project option

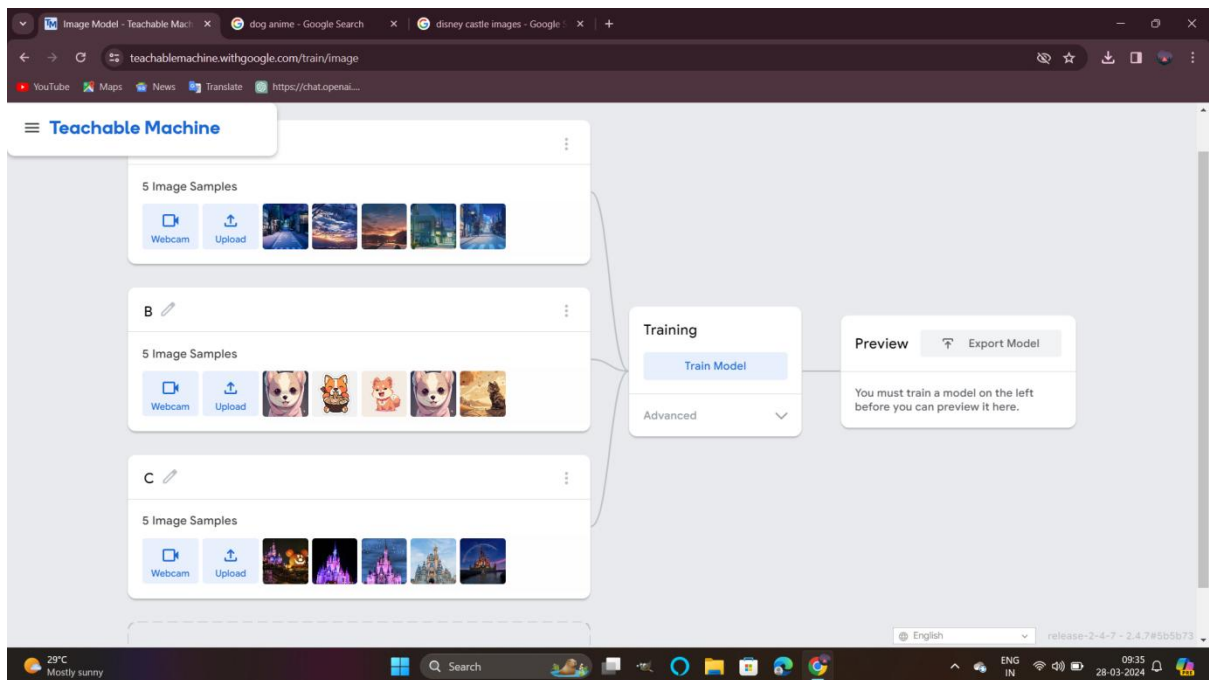


Now, create Machine learning model of two classes namely Class1: **Rohit** Class2: **Geetha and** Class3: **Chanti** respectively (can create more classes, depends on your purpose)

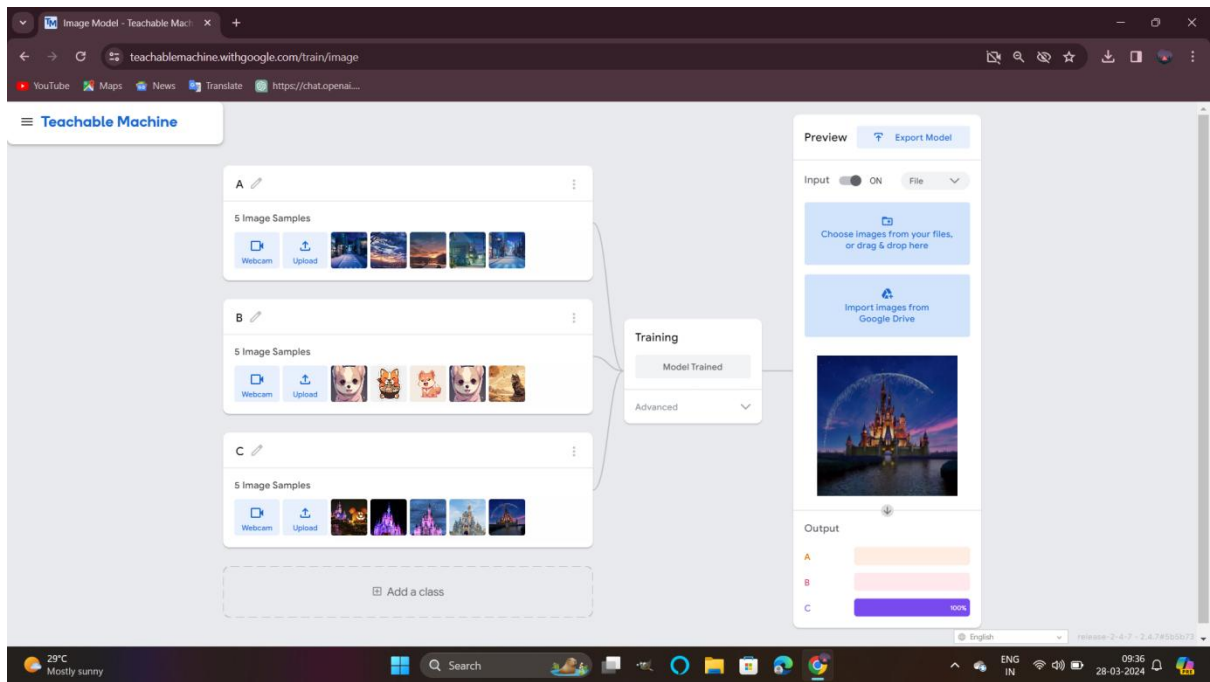
Now, create Machine learning model of two classes namely Class1: **A** Class2: **B and** Class3: **C** respectively (can create more classes, depends on your purpose)



Upload sample images of A,Band C and click on **Train model** finally, to train the model



Now, upload a new input (data point) in the preview panel box, it will predict the output.



MACHINE LEARNING PROGRAMS

Matplotlib:

Data visualization is one of the key skills required for data scientists. There are many tools available for visualizing the data graphically. One such tool for python developers is matplotlib.

Matplotlib is an open-source cross-platform python library used for Data visualization and Graphical plotting.

Installing Matplotlib:

For installing matplotlib in windows, press win+R and type cmd and click enter in the command prompt type, the following command:

Pip install matplotlib

For Linux users, press ctrl+alt+T to open the terminal and type the following command:

Sudo apt-get install python3-matplotlib

Instead of setting up the environment it is recommended to use Jupyter notebook for matplotlib.

import the matplotlib library using the following code.

import matplotlib.pyplot as plt(shortcut name)

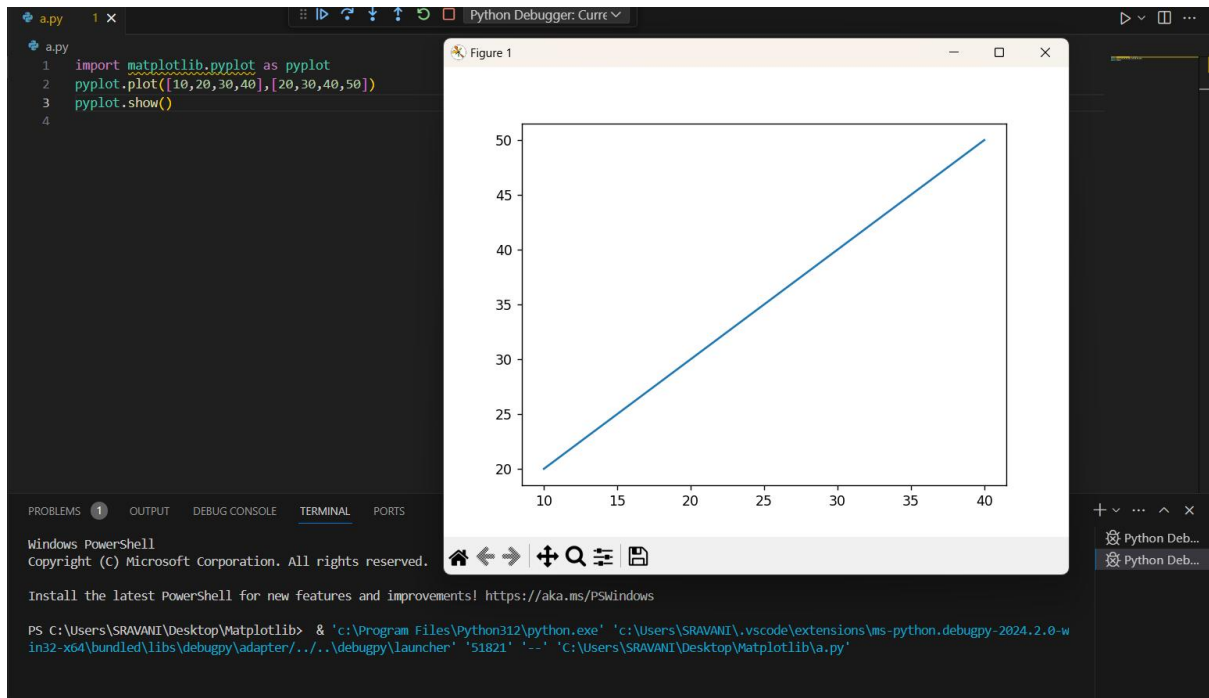
1.creating a simple plot using Matplotlib in python

```
import matplotlib.pyplot as pyplot
```

```
pyplot.plot([10,20,30,40],[20,30,40,50])
```

```
pyplot.show()
```

Output



2.Creating a basic plot with Matplotlib in python with axis labels

```
import matplotlib.pyplot as pyplot

pyplot.plot([10,20,30,40],[20,30,40,50])

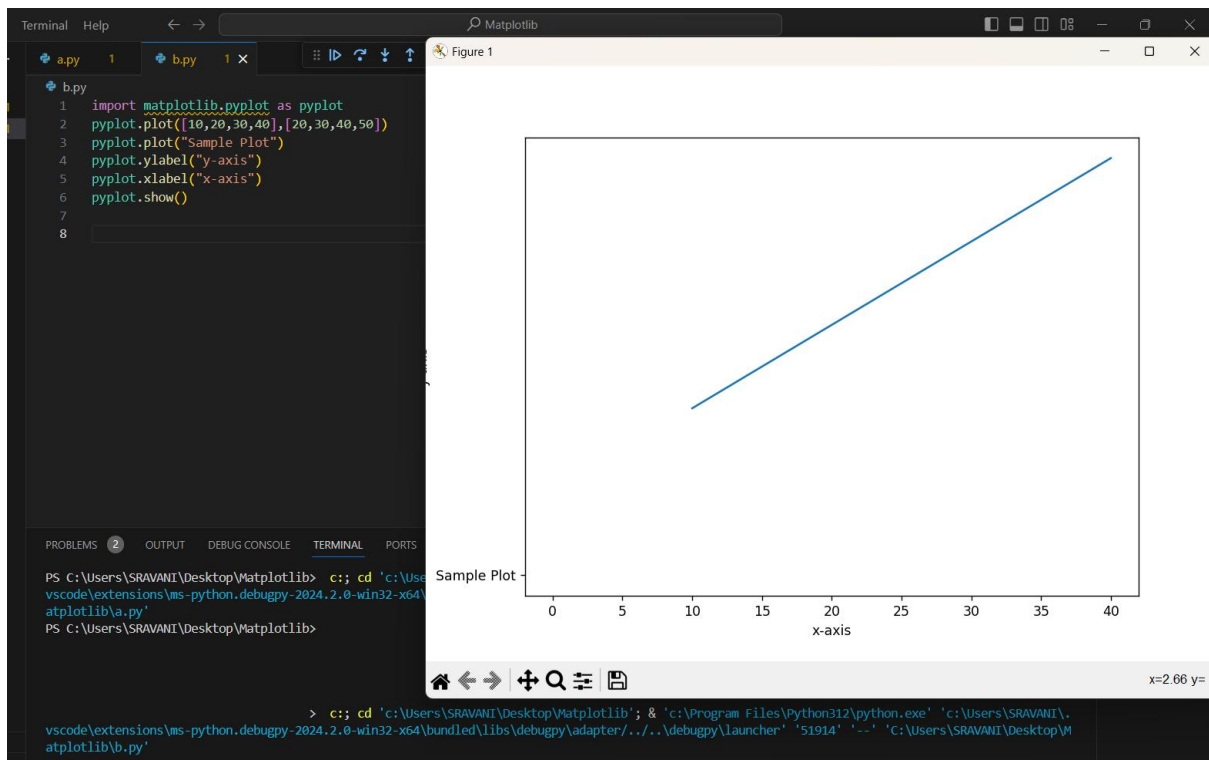
pyplot.plot("Sample Plot")

pyplot.ylabel("y-axis")

pyplot.xlabel("x-axis")

pyplot.show()
```

Output



3. Generating a bar plot in Python with the Data Visualization library

```
import matplotlib.pyplot as pyplot

courses=['Python','Java','C++','C']

values=[24,12,6,18]

pyplot.bar(courses,values,color='Pink')

pyplot.xlabel("Courses")

pyplot.ylabel("Number of students")

pyplot.title("Enrollment")

pyplot.show()
```

Output



4. Generating a scatter plot with Matplotlib in Python

```
import matplotlib.pyplot as pyplot
```

```
x=[2,3,4,6,7,9,8,9,8,12]
```

```
y=[34,65,34,60,83,45,67,87,78,90]
```

```
pyplot.scatter(x,y,color="violet")
```

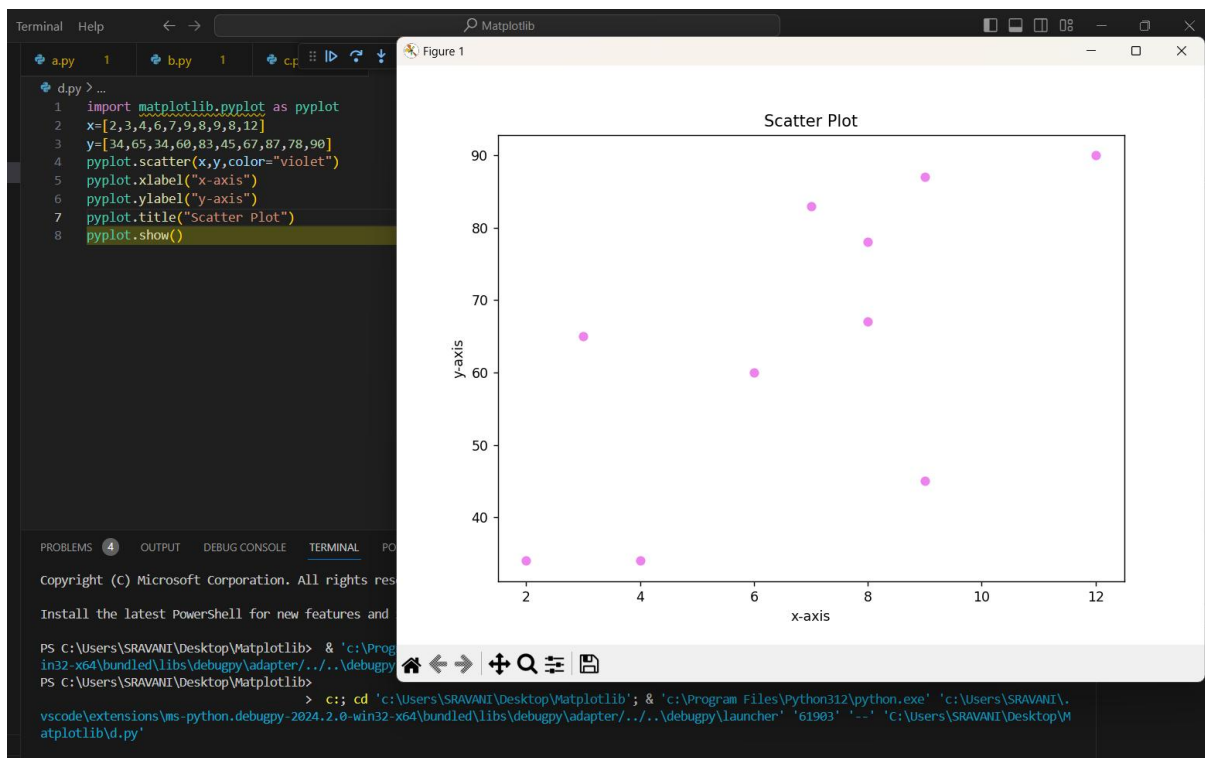
```
pyplot.xlabel("x-axis")
```

```
pyplot.ylabel("y-axis")
```

```
pyplot.title("Scatter Plot")
```

```
pyplot.show()
```

Output



5.Creating a pie chart using matplotlib

```
import matplotlib.pyplot as pyplot
```

```
count=[35,25,25,15]
```

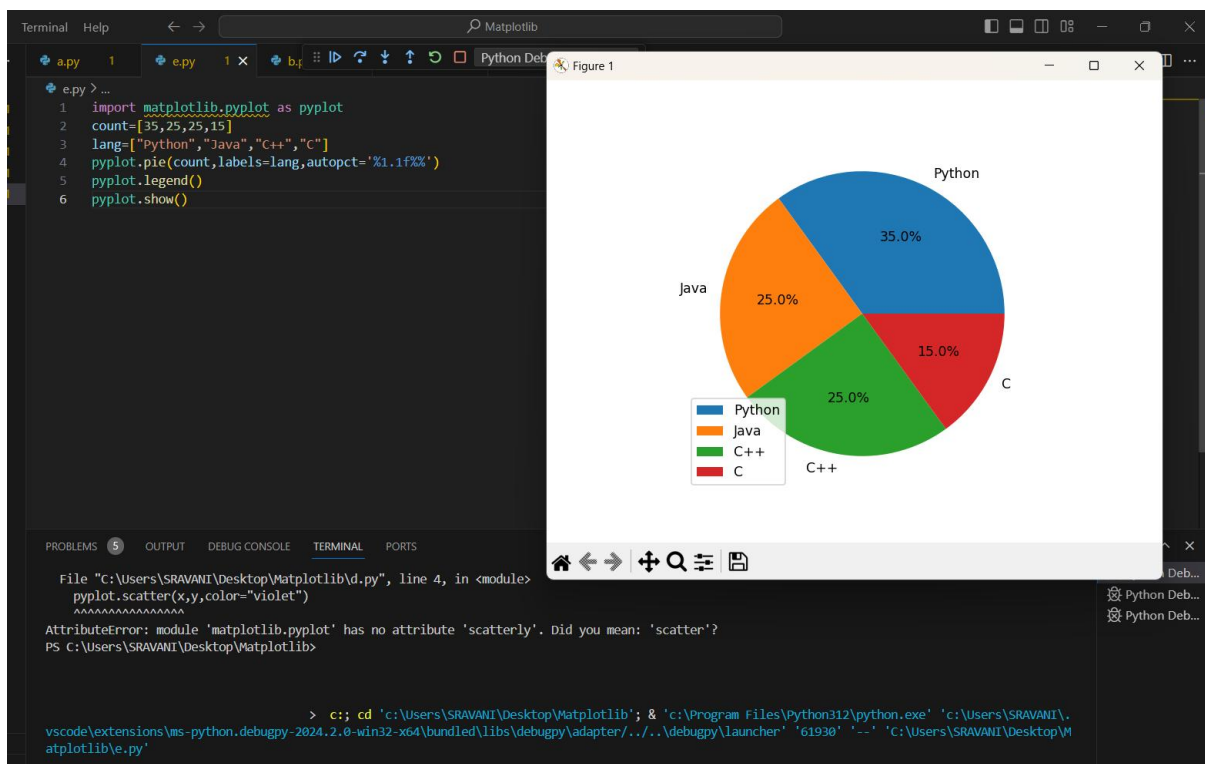
```
lang=["Python","Java","C++","C"]
```

```
pyplot.pie(count,labels=lang,autopct='%1.1f%%')
```

```
pyplot.legend()
```

```
pyplot.show()
```

Output



7. WRITE A PYTHON PROGRAM TO READ CSV DATASET.

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as mlp

data_set=pd.read_csv("placements-dataset.csv")

print(data_set.values)
```

Output

```
PS C:\Users\SRAVANI\Desktop\matplotlib> & 'c:\Users\SRAVANI\AppData\Local\Programs\Python\Python310\python.exe' 'c:\Users\SRAVANI\.vscode\extension
s\ms-python.debugpy-2024.4.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '61645' '--' 'C:\Users\SRAVANI\Desktop\matplotlib\7.py'
[["New York" 6.8 123.0 1]
["Los Angeles" 5.9 106.0 0]
["Chicago" nan 121.0 0]
["New York" 7.4 132.0 1]
["Los Angeles" 5.8 142.0 0]
["Chicago" 7.1 48.0 1]
["New York" 5.7 nan 0]
["Los Angeles" 5.0 63.0 0]
["Chicago" 6.1 156.0 0]
["New York" 5.1 66.0 0]
["Los Angeles" 6.0 45.0 1]
["Chicago" 6.9 138.0 1]
["New York" 5.4 139.0 0]
["Los Angeles" 6.4 116.0 1]
["Chicago" nan 103.0 0]
["New York" 5.1 176.0 0]
["Los Angeles" 5.2 224.0 0]
["Chicago" 3.3 183.0 0]
["New York" 4.0 nan 0]
["Los Angeles" 5.2 132.0 0]
["Chicago" 6.6 120.0 1]
["New York" 7.1 151.0 1]
["Los Angeles" 4.9 120.0 0]
["Chicago" nan 87.0 0]
["New York" 4.7 121.0 0]
["Los Angeles" 5.0 91.0 0]
["Chicago" 7.0 199.0 1]
["New York" 6.0 124.0 1]
["Los Angeles" 5.2 90.0 0]
["Chicago" nan 1.0 0]
["New York" 7.6 128.0 1]
["Los Angeles" 3.9 109.0 0]
["Chicago" 7.0 139.0 1]
["New York" 6.0 149.0 0]
["Los Angeles" 4.8 163.0 0]
["Chicago" 6.8 90.0 1]
["New York" 5.7 140.0 0]
["Los Angeles" 8.1 140.0 1]]
```

8.WRITE A PYTHON PROGRAM TO EXTRACT DEPENDENT VARIABLES FROM DATASET.

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as mlp

data_set = pd.read_csv("placement-dataset.csv")

y=data_set.iloc[:,3].values

print(y)
```

OUTPUT:

```
PS C:\Users\SRAVANI\Desktop\matplotlib> c;; cd 'c:\Users\SRAVANI\Desktop\matplotlib'; & 'c:\Users\SRAVANI\AppData\Local\Programs\Python\Python310\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.4.0-win32-x64\bundle\libs\debugpy\adapter\..\..\debugpy\launcher' '61652' '-
-' 'C:\Users\SRAVANI\Desktop\matplotlib\8.py'
[1 0 0 1 0 1 0 0 0 0 1 1 0 1 0 0 0 0 0 0 1 1 0 0 0 0 1 1 0 0 1 0 1 0 0 1 0
 1 1 0 0 0 1 1 1 1 0 0 1 0 0 0 0 1 1 1 0 1 1 0 0 1 0 1 1 1 1 0 0 1 1 1 1 0
 1 0 0 1 0 0 0 0 1 1 0 1 0 1 0 0 1 1 0 1 0 0 0 1 1 1]
```

9.WRITE A PYTHON PROGRAM TO EXTRACT IN-DEPENDENT VARIABLES FROM DATASET.

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as mlp

data_set = pd.read_csv("placement-dataset.csv")

x=data_set.iloc[:,0:3].values

print(x)
```

OUTPUT:

```
PS C:\Users\SRAVANI\Desktop\matplotlib> c::; cd 'c:\Users\SRAVANI\Desktop\matplotlib'; & 'c:\Users\SRAVANI\AppData\Local\Programs\Python\Python310\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.4.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '61662' '-' 'c:\Users\SRAVANI\Desktop\matplotlib\9.py'
[['New York' 6.8 123.0]
 ['Los Angeles' 5.9 106.0]
 ['Chicago' nan 121.0]
 ['New York' 7.4 132.0]
 ['Los Angeles' 5.8 142.0]
 ['Chicago' 7.1 48.0]
 ['New York' 5.7 nan]
 ['Los Angeles' 5.0 63.0]
 ['Chicago' 6.1 156.0]
 ['New York' 5.1 66.0]
 ['Los Angeles' 6.0 45.0]
 ['Chicago' 6.9 138.0]
 ['New York' 5.4 139.0]
 ['Los Angeles' 6.4 116.0]
 ['Chicago' nan 103.0]
 ['New York' 5.1 176.0]
 ['Los Angeles' 5.2 224.0]
 ['Chicago' 3.3 183.0]
 ['New York' 4.0 nan]
 ['Los Angeles' 5.2 132.0]
 ['Chicago' 6.6 120.0]
 ['New York' 7.1 151.0]
 ['Los Angeles' 4.9 120.0]
 ['Chicago' nan 87.0]
 ['New York' 4.7 121.0]
 ['Los Angeles' 5.0 91.0]
 ['Chicago' 7.0 199.0]
 ['New York' 6.0 124.0]
 ['Los Angeles' 5.2 90.0]
 ['Chicago' nan 1.0]
 ['New York' 7.6 128.0]
 ['Los Angeles' 3.9 109.0]
 ['Chicago' 7.0 139.0]
 ['New York' 6.0 149.0]
 ['Los Angeles' 4.8 163.0]
 ['Chicago' 6.8 90.0]
 ['New York' 5.7 140.0]]
```

10.WRITE A PYTHON PROGRAM TO HANDLE MISSING VALUES IN DATASET.

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as mlp

data_set = pd.read_csv("placement-dataset.csv")

x=data_set.iloc[:,0:3].values

y=data_set.iloc[:,3].values

from sklearn.impute import SimpleImputer

imputer=SimpleImputer(missing_values = np.nan,strategy='mean')

imputer = imputer.fit(x[:,1:3])

x[:,1:3]=imputer.transform(x[:,1:3])

print(x)
```

OUTPUT:

```

PS C:\Users\SRAVANI\Desktop\matplotlib> c:: cd 'c:\Users\SRAVANI\Desktop\matplotlib'; & 'c:\Users\SRAVANI\AppData\Local\Programs\Python\Python310\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.4.0-win32-x64\bundle\libs\debugpy\adapter\..\..\debugpy\launcher' '61667' '-
-' 'C:\Users\SRAVANI\Desktop\matplotlib\10.py'
[ ['New York' 6.8 123.0]
  ['Los Angeles' 5.9 106.0]
  ['Chicago' 5.965217391304349 121.0]
  ['New York' 7.4 132.0]
  ['Los Angeles' 5.8 142.0]
  ['Chicago' 7.1 48.0]
  ['New York' 5.7 117.91666666666667]
  ['Los Angeles' 5.0 63.0]
  ['Chicago' 6.1 156.0]
  ['New York' 5.1 66.0]
  ['Los Angeles' 6.0 45.0]
  ['Chicago' 6.9 138.0]
  ['New York' 5.4 139.0]
  ['Los Angeles' 6.4 116.0]
  ['Chicago' 5.965217391304349 103.0]
  ['New York' 5.1 176.0]
  ['Los Angeles' 5.2 224.0]
  ['Chicago' 3.3 183.0]
  ['New York' 4.0 117.91666666666667]
  ['Los Angeles' 5.2 132.0]
  ['Chicago' 6.6 120.0]
  ['New York' 7.1 151.0]
  ['Los Angeles' 4.9 120.0]
  ['Chicago' 5.965217391304349 87.0]
  ['New York' 4.7 121.0]
  ['Los Angeles' 5.0 91.0]
  ['Chicago' 7.0 199.0]
  ['New York' 6.0 124.0]
  ['Los Angeles' 5.2 90.0]
  ['Chicago' 5.965217391304349 1.0]
  ['New York' 7.6 128.0]
  ['Los Angeles' 3.9 109.0]
  ['Chicago' 7.0 139.0]
  ['New York' 6.0 149.0]
  ['Los Angeles' 4.8 163.0]
  ['Chicago' 6.8 90.0]
  ['New York' 5.7 140.0]

```

11.WRITE A PYTHON PROGRAM TO ENCODE CAREGORICAL VARIABLES.

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as mlp

data_set = pd.read_csv("placement-dataset.csv")

from sklearn.preprocessing import LabelEncoder

label_encoder=LabelEncoder()

data_set['city']=label_encoder.fit_transform(data_set['city'])

print(data_set['city'].values)
```

Output

```
PS C:\Users\SRAVANI\Desktop\matplotlib> c++; cd 'c:\Users\SRAVANI\Desktop\matplotlib'; & 'c:\Users\SRAVANI\AppData\Local\Programs\Python\Python310\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.4.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '55741' '-  
- 'c:\Users\SRAVANI\Desktop\matplotlib\11.py'  
[2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2  
1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1  
0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2]
```


12.DEVELOP A PYTHON PROGRAM FOR EXECUTING THE DATA- PREPROCESSING PHASE WITHIN THE MACHINE LEARNING LIFECYCLE.

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as mlp

from sklearn.preprocessing import LabelEncoder

from sklearn.impute import SimpleImputer

data_set = pd.read_csv("placement-dataset.csv")

data_set['city']=label_encoder.fit_transform(data_set['city'])

x=data_set.iloc[:,0:3].values

y=data_set.iloc[:,3].values

imputer= SimpleImputer(missing_values=np.nan, strategy='mean')

imputer=imputer.fit(X[:,1:3])

print("Categorical data(city variable):")

print(data_set['city'].values)

print("\nIndependent Variables:")

print(x)

print("\nDependent Variables:")

print(y)
```

OUTPUT:

```
PS C:\Users\SRAVANI\Desktop\matplotlib> & 'c:\Users\SRAVANI\AppData\Local\Programs\Python\Python310\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.4.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '55974' '--' 'c:\Users\SRAVANI\Desktop\matplotlib\12.py'
```

Categorical data(city variable):

```
[2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2
 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1
 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 2]
```

Independent Variables:

```
[[ 2.   6.8 123. ]
 [ 1.   5.9 106. ]
 [ 0.   nan 121. ]
 [ 2.   7.4 132. ]
 [ 1.   5.8 142. ]
 [ 0.   7.1  48. ]
 [ 2.   5.7  nan]
 [ 1.   5.   63. ]
 [ 0.   6.1 156. ]
 [ 2.   5.1  66. ]
 [ 1.   6.   45. ]
 [ 0.   6.9 138. ]
 [ 2.   5.4 139. ]
 [ 1.   6.4 116. ]
 [ 0.   nan 103. ]
 [ 2.   5.1 176. ]
 [ 1.   5.2 224. ]
 [ 0.   3.3 183. ]
 [ 2.   4.   nan]
 [ 1.   5.2 132. ]
 [ 0.   6.6 120. ]
 [ 2.   7.1 151. ]
 [ 1.   4.9 120. ]
 [ 0.   nan  87. ]
 [ 2.   4.7 121. ]
 [ 1.   5.   91. ]
 [ 0.   7.   199. ]
 [ 2.   6.   124. ]
 [ 1.   5.2  90. ]
 [ 0.   nan   1. ]
 [ 2.   7.6 128. ]
```

```
[ 2.   6.9  96. ]
 [ 1.   5.   118. ]
 [ 0.   4.   75. ]
 [ 2.   8.5 120. ]
 [ 1.   6.3 127. ]
 [ 0.   6.1  nan]
 [ 2.   7.3 116. ]
 [ 1.   4.9  61. ]
 [ 0.   6.7 154. ]
 [ 2.   4.8 169. ]
 [ 1.   4.9 155. ]
 [ 0.   7.3  50. ]
 [ 2.   6.1  81. ]
 [ 1.   nan   1. ]
 [ 0.   4.9 196. ]
 [ 2.   5.4 107. ]
 [ 1.   6.5  37. ]
 [ 0.   7.5 130. ]
 [ 2.   5.7 169. ]
 [ 1.   5.8 166. ]
 [ 0.   5.1 128. ]
 [ 2.   5.7 132. ]
 [ 1.   4.4 149. ]
 [ 0.   4.9  nan]
 [ 2.   7.3  86. ]
 [ 1.   7.5 158. ]
 [ 0.   5.2 110. ]
 [ 2.   6.8 112. ]
 [ 1.   4.7  52. ]
 [ 0.   4.3 200. ]
 [ 2.   4.4  42. ]
 [ 1.   6.7 182. ]
 [ 0.   6.3 103. ]
 [ 2.   6.2 113. ]]
```

Dependent Variables:

```
[1 0 0 1 0 1 0 0 0 0 1 1 0 1 0 0 0 0 0 0 1 1 0 0 0 0 0 1 1 0 0 1 0 1 0 0 1 0
 1 1 0 0 0 1 1 1 1 0 0 1 0 0 0 0 0 1 1 1 0 1 1 0 0 1 0 1 1 1 1 0 0 1 1 1 1 0
 1 0 0 1 0 0 0 0 1 1 0 1 0 1 0 0 1 1 0 1 0 0 0 1 1 1]
```

13.DEVELOP A PYTHON PROGRAM TO CONCATENATE AND PRINT THE DATA OF INDEPENDENT, DEPENDENT, AND CATEGORICAL VARIABLES, WHILE HANDLING MISSING VARIABLES.

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as mlp

from sklearn.preprocessing import LabelEncoder

from sklearn.impute import SimpleImputer

data_set = pd.read_csv("placement-dataset.csv")

data_set['city']=label_encoder.fit_transform(data_set['city'])

x=data_set.iloc[:,0:3].values

y=data_set.iloc[:,3].values

imputer= SimpleImputer(missing_values=np.nan, strategy='mean')

imputer=imputer.fit(X[:,1:3])

x[:,1:3]=imputer.transform(x[:,1:3])

all_data=np.concatenate((data_set['city'].values.reshape(-1,1),x,y.reshape(-1,1)),axis=1)

print("Categorical Data, Independent Variables, and Dependent Variables for Each Row:")

print(all_data)
```

OUTPUT:

```
PS C:\Users\SRAVANI\Desktop\matplotlib> & 'c:\Users\SRAVANI\AppData\Local\Programs\Python\Python310\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.4.0-win32-x64\bundled\libs\debugpy\adapter\..\..\debugpy\launcher' '56017' '--' 'C:\Users\SRAVANI\Desktop\matplotlib\13.py'
```

Categorical Data, Independent Variables, and Dependent Variables for Each Row:

```
[[ 2.      2.      6.8      123.      1.      ]
 [ 1.      1.      5.9      106.      0.      ]
 [ 0.      0.      5.96521739 121.      0.      ]
 [ 2.      2.      7.4      132.      1.      ]
 [ 1.      1.      5.8      142.      0.      ]
 [ 0.      0.      7.1      48.       1.      ]
 [ 2.      2.      5.7      117.91666667 0.      ]
 [ 1.      1.      5.       63.       0.      ]
 [ 0.      0.      6.1      156.      0.      ]
 [ 2.      2.      5.1      66.       0.      ]
 [ 1.      1.      6.       45.       1.      ]
 [ 0.      0.      6.9      138.      1.      ]
 [ 2.      2.      5.4      139.      0.      ]
 [ 1.      1.      6.4      116.      1.      ]
 [ 0.      0.      5.96521739 103.      0.      ]
 [ 2.      2.      5.1      176.      0.      ]
 [ 1.      1.      5.2      224.      0.      ]
 [ 0.      0.      3.3      183.      0.      ]
 [ 2.      2.      4.       117.91666667 0.      ]
 [ 1.      1.      5.2      132.      0.      ]
 [ 0.      0.      6.6      120.      1.      ]
 [ 2.      2.      7.1      151.      1.      ]
 [ 1.      1.      4.9      120.      0.      ]
 [ 0.      0.      5.96521739 87.       0.      ]
 [ 2.      2.      4.7      121.      0.      ]
 [ 1.      1.      5.       91.       0.      ]
 [ 0.      0.      7.       199.      1.      ]
 [ 2.      2.      6.       124.      1.      ]
 [ 1.      1.      5.2      90.       0.      ]
 [ 0.      0.      5.96521739 1.       0.      ]
 [ 2.      2.      7.6      128.      1.      ]
 [ 1.      1.      3.9      109.      0.      ]
 [ 0.      0.      7.       139.      1.      ]
 [ 2.      2.      6.       149.      0.      ]
 [ 1.      1.      4.8      163.      0.      ]
 [ 0.      0.      6.8      90.       1.      ]]
```

14.DEVELOP A PYTHON PROGRAM TO SPLIT A DATASET INTO TRAINING AND TESTING SETS.

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as mlp

from sklearn.preprocessing import LabelEncoder

from sklearn.model_selection import train_test_split

from sklearn.impute import SimpleImputer

data_set = pd.read_csv("placement-dataset.csv")

data_set['city']=label_encoder.fit_transform(data_set['city'])

x=data_set.iloc[:,0:3].values

y=data_set.iloc[:,3].values

imputer= SimpleImputer(missing_values=np.nan, strategy='mean')

x[:,1:3]=imputer.fit_transform(x[:,1:3])

x_train, x_test, y_train, y_test =train_test_split(x,y,test_size=0.2,random_state=0)

print("Shape of x_train:", x_train.shape)

print("Shape of x_test:", x_test.shape)

print("Shape of y_train:", y_train.shape)

print("Shape of y_test:", y_test.shape)
```

OUTPUT:

```
PS C:\Users\SRAVANI\Desktop\matplotlib> c;; cd 'c:\Users\SRAVANI\Desktop\matplotlib'; & 'c:\Users\SRAVANI\AppData\Local\Programs\Python\Python310\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.4.0-win32-x64\bundle\libs\debugpy\adapter\..\..\debugpy\launcher' '56022' '-
-' 'c:\Users\SRAVANI\Desktop\matplotlib\14.py'
Shape of x_train: (80, 3)
Shape of x_test: (20, 3)
Shape of y_train: (80,)
Shape of y_test: (20,)
```

15.WRITE A PYTHON PROGRAM IMPLEMENTING STANDARDIZATION TECHNIQUE IN FEATURE SCALING PROCESS.

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as mlp

from sklearn.preprocessing import LabelEncoder,StandardScaler

from sklearn.model_selection import train_test_split

from sklearn.impute import SimpleImputer

data_set = pd.read_csv("placement-dataset.csv")

label_encoder=LabelEncoder()

data_set['city']=label_encoder.fit_transform(data_set['city'])

x=data_set.iloc[:,0:3].values

y=data_set.iloc[:,3].values

imputer= SimpleImputer(missing_values=np.nan, strategy='mean')

x[:,1:3]=imputer.fit_transform(x[:,1:3])

x_train, x_test, y_train, y_test= train_test_split(x,y,
    test_size=0.2,random_state=0)

scaler=StandardScaler()

x_train_scaled=scaler.fit_transform(x_train)

x_test_scaled=scaler.transform(x_test)

print("strandardized training features (x_train_scaled):\n", x_train_scaled)

print("\nstrandardized training features (x_test_scaled):\n", x_test_scaled)
```

```

print("\n mean of standardized training features (x_train_scaled):",
      np.mean(x_train_scaled, axis=0))

print("starndard deviation of standardized training features (x_train_scaled):",
      np.std(x_train_scaled, axis=0))

print("\n mean of standardized training features (x_test_scaled):",
      np.mean(x_test_scaled, axis=0))

print("starndard deviation of standardized training features (x_test_scaled):",
      np.std(x_test_scaled, axis=0))

mean_train_rounded=np.around(np.mean(x_train_scaled, axis=0))

std_train_rounded=np.around(np.std(x_train_scaled,axis=0), decimals=0)

mean_test_rounded=np.around(np.mean(x_test_scaled, axis=0))

std_test_rounded=np.around(np.std(x_test_scaled,axis=0), decimals=0)

print("\n mean of standardized training features
      (x_train_scaled):",mean_train_rounded)

print("starndard deviation of standardized training features
      (x_train_scaled):",std_train_rounded)

print("\n mean of standardized training features
      (x_test_scaled):",mean_test_rounded)

print("starndard deviation of standardized training features
      (x_test_scaled):",std_test_rounded)

```

OUTPUT:

```
PS C:\Users\SRAVANI\Desktop\matplotlib> & 'c:\Users\SRAVANI\AppData\Local\Programs\Python\Python310\python.exe' 'c:\Users\SRAVANI\.vscode\extensions\ms-python.debugpy-2024.4.0-win32-x64\bundle\libs\debugpy\adapter\..\..\debugpy\launcher' '56039' '--' 'c:\Users\SRAVANI\Desktop\matplotlib\15.py'
```

```
standardized training features (x_train_scaled):  
[[-1.53591893e-02 -8.30719549e-03 5.93170699e-01]  
 [-1.24409433e+00 2.29669522e-02 -2.59945119e-01]  
 [ 1.21337595e+00 1.28175140e+00 3.96297818e-01]  
 [-1.24409433e+00 1.12880127e-01 8.82282171e-02]  
 [ 1.21337595e+00 2.29669522e-02 -1.04743664e+00]  
 [ 1.21337595e+00 5.62446000e-01 5.27546405e-01]  
 [ 1.21337595e+00 -2.46772572e-01 8.82282171e-02]  
 [ 1.21337595e+00 2.02793302e-01 -1.93227087e-02]  
 [-1.53591893e-02 4.72532826e-01 -1.68180482e+00]  
 [-1.53591893e-02 -9.66077970e-01 8.99417403e-01]  
 [ 1.21337595e+00 -8.30719549e-03 -2.46929634e+00]  
 [-1.24409433e+00 -9.66077970e-01 1.79628275e+00]  
 [ 1.21337595e+00 1.19183822e+00 -6.09941352e-01]  
 [-1.24409433e+00 -1.77529654e+00 -8.50563762e-01]  
 [ 1.21337595e+00 -1.05599114e+00 5.93170699e-01]  
 [ 1.21337595e+00 2.29669522e-02 2.21299702e-01]  
 [ 1.21337595e+00 -1.77529654e+00 8.82282171e-02]  
 [-1.24409433e+00 1.12880127e-01 -1.06931141e+00]  
 [ 1.21337595e+00 2.92706476e-01 3.08798760e-01]  
 [-1.24409433e+00 6.52359175e-01 8.77542639e-01]  
 [-1.53591893e-02 -6.69462225e-02 -1.72446061e-01]  
 [-1.53591893e-02 1.19183822e+00 5.05671641e-01]  
 [ 1.21337595e+00 1.46157775e+00 -5.44317058e-01]  
 [-1.24409433e+00 -5.16512096e-01 2.55205587e-03]  
 [-1.53591893e-02 -1.56859397e-01 6.15045464e-01]  
 [ 1.21337595e+00 -7.86251620e-01 1.35878746e+00]  
 [-1.24409433e+00 -2.40468876e+00 1.51191081e+00]  
 [-1.53591893e-02 -9.66077970e-01 4.40047347e-01]  
 [-1.24409433e+00 4.72532826e-01 1.00879123e+00]  
 [-1.24409433e+00 1.01201187e+00 -1.44118241e+00]  
 [-1.53591893e-02 1.37166457e+00 9.65041697e-01]  
 [-1.24409433e+00 -1.05599114e+00 -4.11974733e-02]  
 [ 1.21337595e+00 7.42272350e-01 1.99424937e-01]  
 [-1.53591893e-02 -1.05599114e+00 1.07441552e+00]  
 [-1.53591893e-02 -6.96338445e-01 -5.22442294e-01]  
 [-1.24409433e+00 -2.22486242e+00 2.60564904e+00]
```


MINI PROJECT

1.Create a responsive blog layout with multiple articles.How will you use media queries to ensure the blog layout adapts to different screen sizes.

BLOG LAYOUT

Blog layout refers to the way a blog is organized and presented to readers. It includes elements like the header, navigation menu, sidebar, content area, and footer. A well-designed blog layout makes it easy for readers to navigate and find the information they're looking for. Creating a responsive blog layout with multiple articles involves designing a layout that adjusts smoothly across various screen sizes while accommodating multiple pieces of content.

MEDIA QUERIES

Media queries are a fundamental component of responsive web design. They allow developers to apply CSS styles based on various characteristics of the device or browser viewport, such as screen size, orientation, resolution, and more. Media queries enable websites to adapt their layout and appearance to different devices, ensuring optimal user experience across desktops, tablets, smartphones, and other devices.To ensure the blog layout adapts to different screen sizes, we can use media queries in CSS to adjust the layout based on the device's screen width

HTML CODE

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<link rel="stylesheet" href="M.css" />

<title>Responsive Blog Layout</title>
```

</head>

<body>

<div class="container">

<div class="article">

<h2>Fruits and Vegetables</h2>

<p>The term “fruit” has several meanings. To a botanist, it means the part of a plant that contains seeds. According to this definition, fruits include most nuts, as well as vegetables, such as cucumbers and tomatoes.

To most of us, though, “fruit” is defined as the soft, edible, seed-bearing part of a perennial plant. A perennial is a plant that lives for more than one growing season. Fresh fruits are rich in carbohydrates, vitamins, minerals, and fiber. They can be preserved by freezing, canning, or drying.

Different fruits grow in different climates, and may not grow well in climates that are too cold, hot, dry, or wet. Based on the kind of climate in which they grow, fruits can be classified in different groups: temperate fruits, subtropical fruits, and tropical fruits.

Temperate fruits grow best where there is a well-defined cold season, as in the U.S. states of Washington and Oregon. Temperate fruits include apples, berries, grapes, pears, plums, and peaches.

</p>

</div>

<div class="article">

<h2>Milk and Dairy</h2>

<p>Much of the world's milk, cream, butter, yogurt, and cheese come from dairy cows. However, goats, camels, reindeer, sheep, yaks, and water buffalo supply milk products as well.

In much of Asia, people have traditionally consumed "milk" made from soybeans. Soy milk is made from soaking and grinding soybeans with water. Soy milk contains about the same protein as cow's milk.

Milk and dairy products supply proteins, carbohydrates, fats, and essential vitamins and minerals. The Maasai people of East Africa use milk as a staple food, and it constitutes a huge part of their diet. Yogurt, a fermented milk product, is a staple food in parts of the Middle East and India.</p>

</div>

<div class="article">

<h2>Legumes and cereals</h2>

<p>Legumes are plants that are raised for their edible seeds or seed pods. Peas, lima beans, soybeans, peanuts, and lentils are all legumes. Legumes that are harvested for their dry seeds, such as beans or lentils, are called pulses. Pulses are food staples in India and Pakistan.

Legumes and pulses contain high amounts of protein. In addition, they supply iron, other minerals, and vitamins.

The term “meat” usually refers to the edible flesh of mammals, such as cattle, pigs, and sheep. Meat is a high-protein food and is rich in other nutrients, as well.

The United States, Australia, Russia, and Argentina raise much of the world’s beef cattle. Beef can come from cows, bulls, or steers. Cows are adult female cattle. Bulls are adult male cattle capable of reproduction. Steers are adult male cattle that have been castrated, or been made incapable of reproduction. Meat from very young cattle, or calves, is called veal.</p>

</div>

</div>

</body>

</html>

CSS CODE

body {

font-family: 'Trebuchet MS', 'Lucida Sans Unicode', 'Lucida Grande', 'Lucida Sans', Arial, sans-serif;

background-color: blueviolet;

}

.container {

max-width: 1200px;

margin: 0 auto;

```
padding: 20px;

display: flex;

}

.article {

width: 30%;

margin-bottom: 20px;

background-color: whitesmoke;

padding: 20px;

border-radius: px;

}

.article img {

max-width: 50%;

height: fit-content;

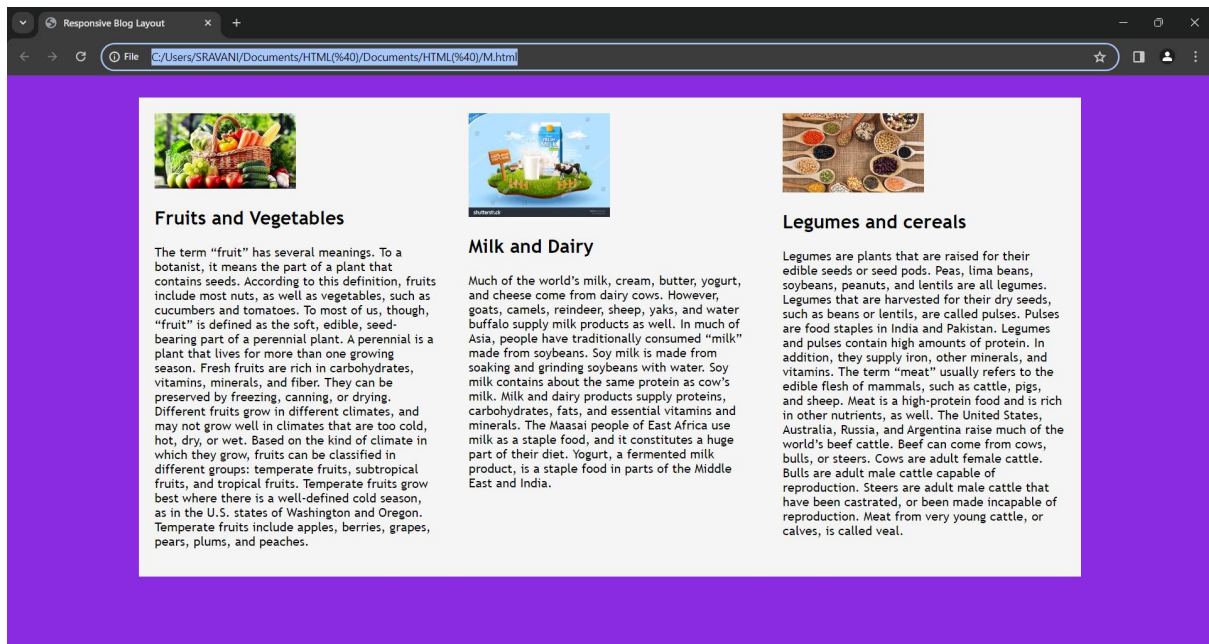
}

@media only screen and (min-width: 601px) and (max-width: 1024px) {

}

}
```

OUTPUT



Fruits and Vegetables

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Milk and Dairy

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Legumes and cereals

Legumes are plants that are raised for their edible seeds or seed pods. Peas, lima beans, soybeans, peanuts, and lentils are all legumes. Legumes that are harvested for their dry seeds, such as beans or lentils, are called pulses. Pulses are food staples in India and Pakistan. Legumes and pulses contain high amounts of protein. In addition, they supply iron, other minerals, and vitamins. The term "meat" usually refers to the edible flesh of mammals, such as cattle, pigs, and sheep. Meat is a high-protein food and is rich in other nutrients, as well. The United States, Australia, Russia, and Argentina raise much of the world's beef cattle. Beef can come from cows, bulls, or steers. Cows are adult female cattle. Bulls are adult male cattle capable of reproduction. Steers are adult male cattle that have been castrated, or been made incapable of reproduction. Meat from very young cattle, or calves, is called veal.

